Building body confidence in women through shopping experience

MA Degree Project

IxD Interaction Design Estonian Academy of Arts

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I dedicate this work to:

all brilliant women, who's advancement was undermined by 'beauty myth'

all women questioning the 'beauty myth'

good people of DUX, who brought out the best in me

Riina Raudne I look up to

Tanel Kärp and Nesli Hazal Akbulut, giving me creative freedom

Abstract

People’s concern about their body image has never been more serious, and yet the set ideals have never been so unattainable.

Through research I discovered that shopping is the main body dissatisfaction factor affecting women. Therefore I aimed at not only providing women with a practical solution of effective size search but also help them reduce their body dissatisfaction.

My solution offers an alternative way of shopping by creating location-based micro-communities of shoppers sharing similar body traits and showing relevant reviews in real-time context through augmented reality.

On psychological level it provides means of resisting shame by practicing critical awareness, reaching out to other women and speaking up about shame in the context of shopping.

Kokkuvõte

Inimeste mure oma keha kuvandi osas pole kunagi olnud nii suur, siiski on seatud ideaalid aga kättesaamatud.

Uurimistöö käigus avastasin, et ostmine on peamine keha rahulolematuse tegur, mis naisi mõjutab. Sellele tuginedes seadsin eesmärgi mitte disainida ainult efektiivne suuruse otsingu lahendus, vaid samuti aidata neil vähendada rahulolematust oma keha üle.

Minu lahendus pakub alternatiivset ostlemisviisi, luues asukoha põhiseid mini keskkondi, milles saavad sarnase kehatüübiga ostlejad jagada usaldusväärseid arvustusi reaalajas, kasutades liitreaalsust.

Psühholoogilisel tasandil pakub lahendus tuge häbile vastu seismiseks, rakendades kriitilist teadlikkust, pöördudes teiste naiste poole ja rääkides häbist poodlemise kontekstis.

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1. Introduction

1.1 Background

Due to the vast influence of social media and dominance of visual content across the majority of channels of communication, people's concern about their body image has never been so serious, and yet the set ideals have never been so unattainable.

According to Mental Health Foundation survey [1], carried out in 2019 in the UK, only 21% of adults (23% of men and 19% of women) felt satisfied with their body image, 20% (15% of men and 25% of women) felt shame and 34% (25% of men and 43% of women) felt low, and 19% (12% of men and 26% of women) felt disgusted because of their body image. In Mail.Ru survey, carried out in 2017 in Russia [2], 70% of women and 63% of men admitted being dissatisfied with their body. Based on World Health Organization data [3] published in 2015, 38% of 15-year-old girls and 22% of boys in Estonia think that they are too fat, although same statistics indicated that the real percentage of overweight and obese people in this age group was 9% among females and 19% among males [4].

Body dissatisfaction has become a global tendency, but women have higher rates of body image concern, that can result in serious mental health problems, such as depression, anxiety, eating disorders and influence all spheres of a person's life.

1.2 Personal interest

For my thesis project, I got inspired by Naomi Wolf, who in her book 'Beauty Myth' [5], explored the concept of beauty as a social currency, and stated that 'beauty myth and attainment of beauty is a new virtue that is well suited for a new consumer imperative and a new justification for economic unfairness in the workplace. Somehow, somewhere, someone must have figured out that women will buy more things if they are kept in the self-hating, ever-failing, hungry, and sexually insecure state of being aspiring "beauties."

It started me thinking that all women to a certain degree participate in mass culture that is imposed by media and advertisement, see false images selling certain lifestyles and are shamed into buying things. Even female magazines as a part of mass culture have no right to write the way they want, because they will lose lots of advertisers. This led me to the idea of finding a solution for adjusting current situation in a way that it could promote body confidence and realistic standards.

1.3 Project goals

My **goal for this project** is to explore factors influencing body satisfaction/dissatisfaction in women and develop a solution that can help women resist social pressure and build their body confidence. To achieve this, I will:

1. find opportunities for how to adjust current paradigm to the needs of women
2. sketch and design concept
3. test my prototype with my target audience and iterate on it

My **personal goal** with this project is to apply the acquired knowledge during the course, in particular:

- improve my skills of interviewing people and analyzing the results
- learn how to conduct usability testing using specialized tools
- explore new technologies that can add to my solution.

2. Design Approach

I followed the Double Diamond framework, developed by Design Council in 2005. The two diamonds represent a process of exploring an issue more widely or deeply (divergent thinking) and then taking focused action (convergent thinking)[6]. Using The Double Diamond, I structured my project in four phases:

- **Discover,** where I started with the initial question of how to build body confidence in women, looked for the problem to solve by using the collage method, doing user interviews and carrying out desk research.
- **Define,** where I made sense of my research by analyzing user interviews, generating opportunities and formulating How Might We Questions.
- **Develop,** where I looked for potential solutions by brainstorming and sketching, evaluated my ideas and picked a concept for further development.
- **Deliver,** where I created prototypes and tested them with potential users.

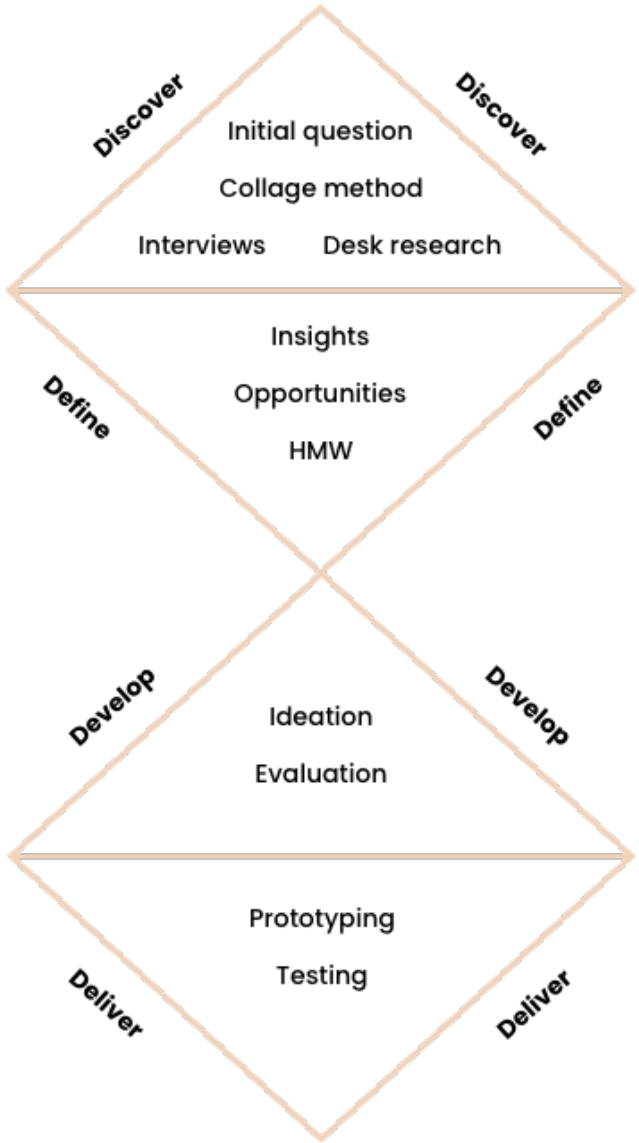


Figure 1: My adaptation of Double Diamond

3 Key definitions and focus

3.1 Body image

Body image is defined as the picture an individual forms in their mind that constitutes beliefs, feelings, sensations, behaviours and self-perception, concerning their own body [7]. It also includes experiences of embodiment and investment in appearance. **Body dissatisfaction** can be defined as a negative evaluation of body size, shape, muscle tone, weight, and it usually involves a perceived difference between a person's assessment of their body and their ideal body.

3.2 Target audience

When defining my target audience for the project, I found out that same body dissatisfaction concerns are assigned to women from late teens till late 30s. And only middle-aged women (40–49) were reported as having more realistic age-based comparisons, being concerned to "look good for their age," rather than to look like women in magazine pictures or on television, seeing body-related changes such as weight gain as an inevitable[8]. That's why I decided to see what happens in the mid-period and design for women between 27 - 35, the age when women most likely have already graduated from university and have less peer pressure, have a career or family life, in other words, changed their lifestyle from the one they had when they were younger.

3.3 Focus areas

To define my focus in this broad and complex topic, I have reviewed existing models of body image formation and picked the one that provided the most extensive overview and could help me define what parts I could research and create impact on as a designer.

Cognitive-behavioural model

According to cognitive-behavioural model [9], body image is influenced by events from the past and present (figure 2).

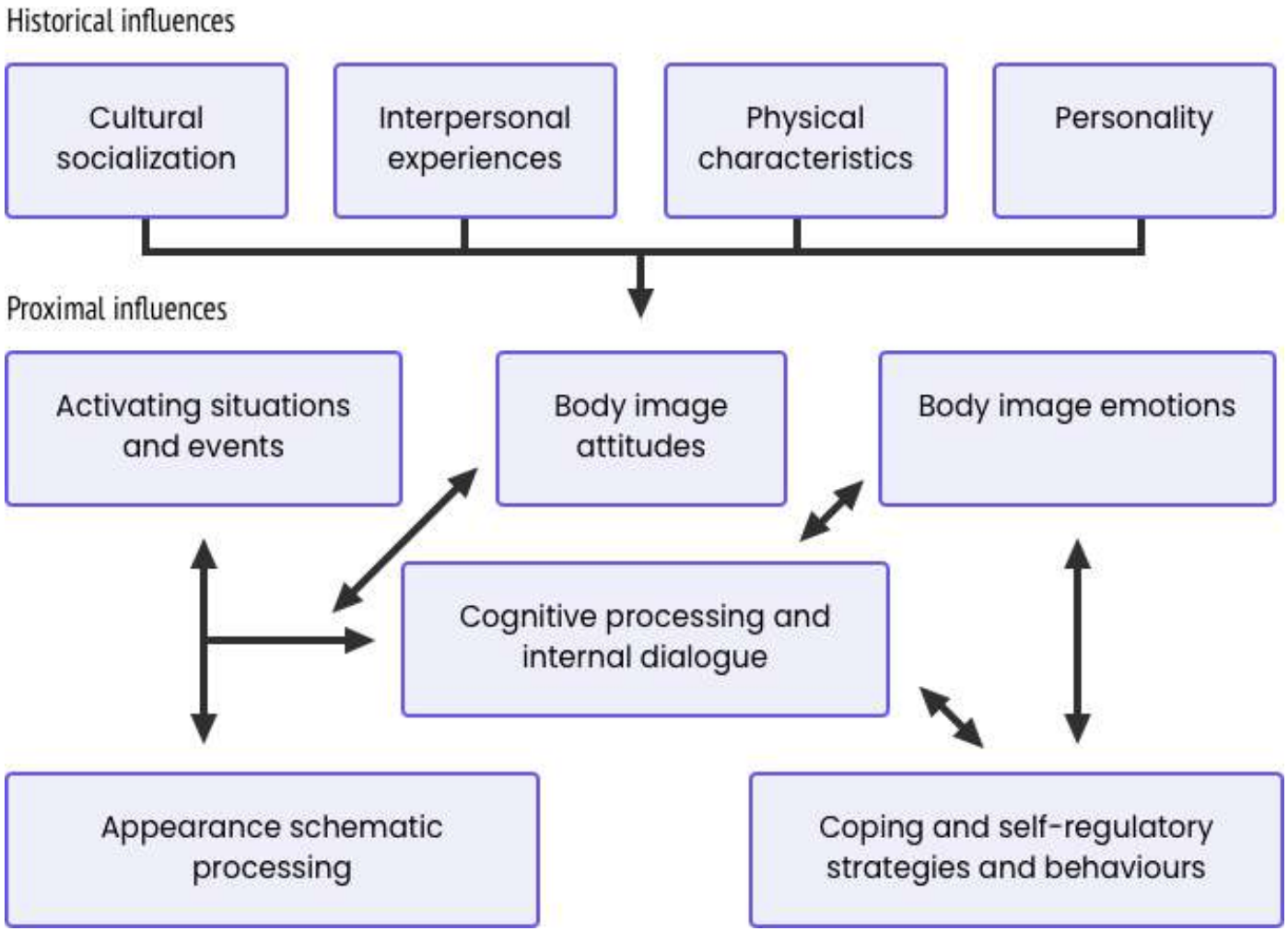


Figure 2: Cognitive-behavioral model

Interpersonal influences manifest themselves in our relations with people, especially peers, family or romantic partners. Parents can influence the body image of their children by commenting about a child's weight or appearance, introducing particular food rules or by setting an example through dieting and overexercising.

Peers influence women through "fat talk" (discussion about their and other people's weight loss or weight gain), comments and conversations about weight and shape, and the belief that popularity is dependent on being thin, or having an 'idol' to copy.

Cultural socialization implies that people learn from an early age how society values different physical characteristics and what meanings are assigned to them. For example, if we think about what a superhero might look like, we will probably imagine a muscular man.

Physical characteristics are connected with the fact that a body is dynamic and changes over time. Even if puberty, facial acne or extra weight were a matter of an individual's past, body image could still remember them. That' why, for example, people who lost weight might still feel fat.

Personality traits such as perfectionism or insecure attachment patterns can be catalysts to body dissatisfaction.

Everyday situations and events trigger certain emotions, thoughts and feelings, and some of them can be painful and disturbing, and initiate a chain of responses. Based on those events, an individual can formulate certain assumptions about themselves, and starts performing certain actions, aimed at appearance fixing.

All emotions, comments from people, assumptions make foundation for **schemas**, which determine how an individual perceives reality, thinks about events and themselves.

3.4 Thesis direction

Using the cognitive -behavioural model, I reviewed solutions that are already suggested to people and found out that there are solutions that:

- deal with cognitive and emotional part, for example, Perspectives app that offers digital cognitive behavioural therapy by offering simple exercises to help with body image concerns,

- offer coping strategies, for example, Headspace teaches how to practice mindfulness for body confidence,

- influence cultural socialization and interpersonal experiences. For example, Dove Self Esteem Project hosts resources for teachers and parents to help them raise body confident and critically aware children,

- some solutions address body schema using virtual reality to reduce psychological body size overestimation.

I didn't find solutions focused on everyday situations and events that cause body dissatisfaction and decided that during a user research I could find such situation and develop a solution for it. In my opinion addressing everyday situations can have a higher impact on women and provide a more continuous support, and prevent women from creating distractive and negative assumptions about themselves.

4. User research

4.1 Collage Session

I decided to use the collage method as a starting point of my research for three reasons. Firstly, it could provide me with a way to narrow down without defining any specific questions. Secondly, images are powerful triggers of unconscious emotions and past experiences, and it could be at hand with such a personal and very emotional topic as relations with the body. Thirdly, it could help me understand how sensitive women are about it and how careful I should be during interviews.

For the collage session I invited 3 women matching my target audience:

- Kate, school teacher, 32,
- Helen, 30, jewellery maker,
- Ann, 28, QA engineer.

I gave them different types of magazines, such as Vogue, Cosmopolitan, Shape and National Geographic. I suggested to tear out all the images they like and put them in a stack, and only then start creating collages. Before the collage session, I created my variant of collage to check how my ideas and expectations align with others or how different they are. Also, it could serve me as a reminder of confirmation biases all people have (Figure 3) and stop me from taking a direction I want to go instead of focusing on what women are experiencing.

After collage session (Figure 4), I asked all women to tell about their choice of images. Most women admitted that body confidence has a lot to do with clothes, and my collage was very different compared to theirs.

I singled out the **themes**, that can have further development during interviews:

- clothes as a part of being body confident,
- expressing yourself and being brave and showing yourself to the world,
- receiving attention from men.

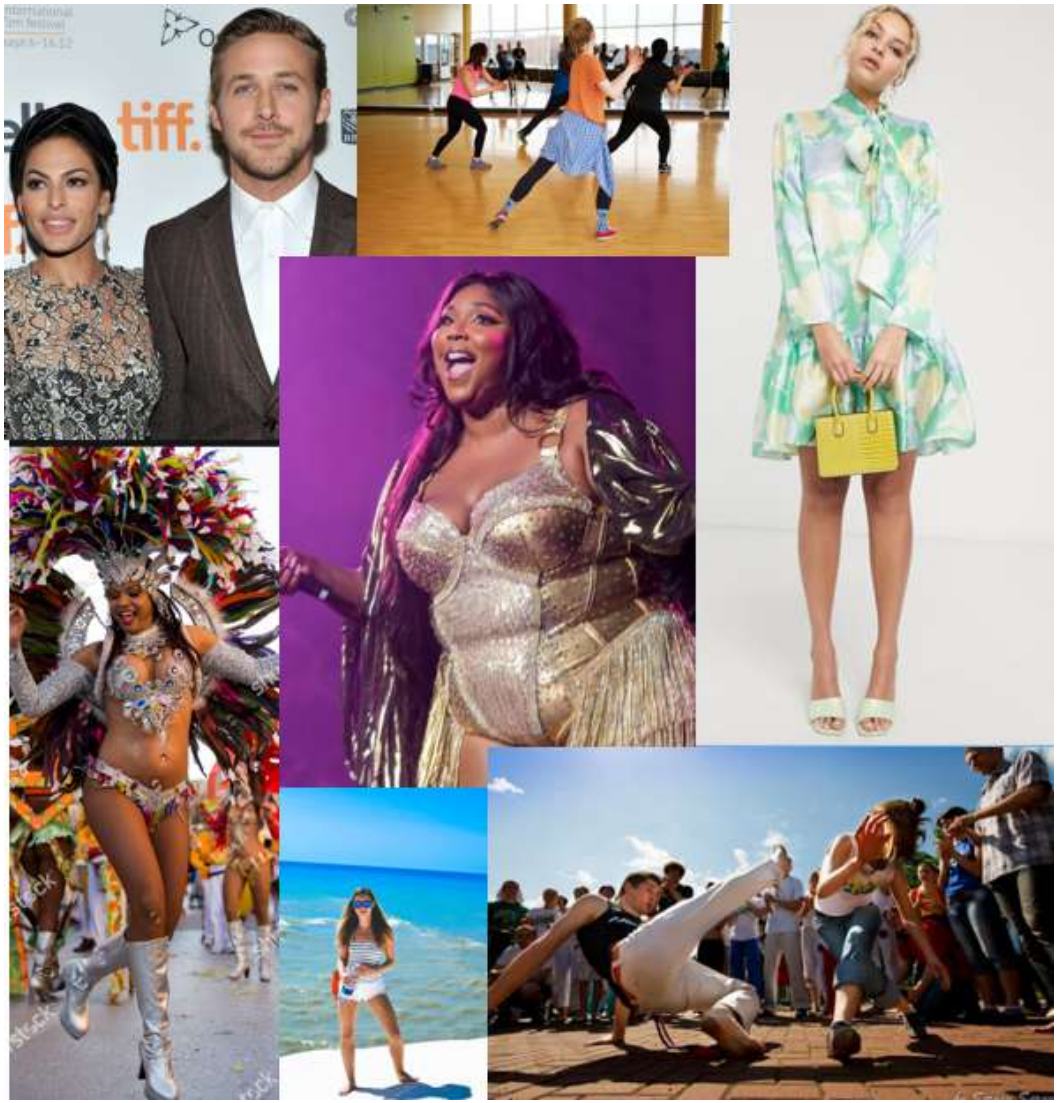


Figure 3: My collage about body

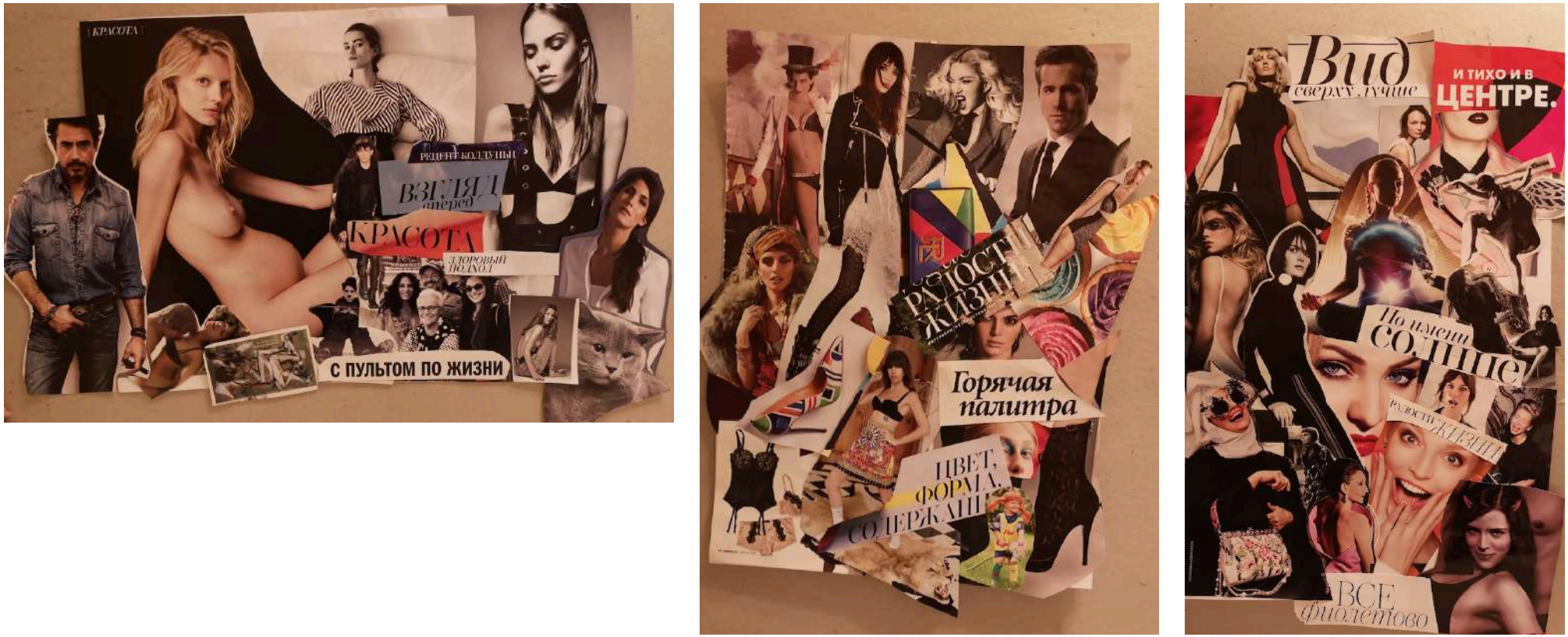


Figure 4: Collages made by participants

4.2 User Interviews

I conducted eight interviews (Figure 5), of women matching my age, profession and background requirements:

- age between 27 - 35
- university graduates
- having a career

I chose the form of an unstructured interview, and prepared some questions in advance, but tried to keep it informal and free-flowing, so that my interviewees could feel as if they are having a normal conversation. I started with broader questions and narrowed down when I saw a recurring pattern in women's stories.

Main goals of the interviews:

- find out what everyday situations and events trigger negative emotions, and what emotions and thoughts women experience during those events and situations,
- learn about coping techniques women worked out in response to activating events,
- discover common values among women, that can be supported in the development of my concept,
- see if themes discovered during the collage session have any development in what women were telling.

4.3 Interview results

When I analyzed the interviews, I found out that women worked out specific ways how to cope with body dissatisfaction through fitness, finding people they feel support from, hobbies or raising their critical awareness of the social phenomena connected with body image.

Fitness is for mental health, not for 'beach body'.

Women see fitness as a way of coping with body dissatisfaction, and use it together with other practices to feel better about themselves.

'Yoga and other body practices help me a lot, because they teach me that I am my own source of happiness.'



Figure 5: Women interviewed for the project

‘Fitness goals should be achievable. Even if I do a bit better than yesterday, I feel proud of myself.’

‘I was depressed because of the weight gain, and one day I decided not to judge myself on how I look, because then I will never be happy until I get there. It is a vicious circle. I did pick up random stuff like art and horse riding. I didn't use makeup. Tried to get used to my own face because I felt ugly.’

‘In the gym you're making your attempt to make your life better. And people know that.'

‘I listed down my achievements and focused on them. ‘

Critical awareness is already in a toolbox.

Women are critical about what is shown in media and realize how destructive the current body ideal is. They look back at their past and regret being influenced by those ideals when they were younger.

‘I don't see normal bodies, representing us. You know, only the skinny type that affects a lot of especially young girls because that's what they see and that's what they strive to be. And that’s a kind of the idealized image that they see.'

‘Christina Aguilera and Britney Spears were very popular. Flat bellies, open clothes. I saw them and I was very much jealous of how they looked. Now I understand that I wasn't even mature enough to look the way that they did.’

‘I follow FB group How Humans Look, and I noticed that no matter what a person looks like, plus size, fitness goddess, or a model looking nymph, there will be some comments about their ugliness.’

Men are not the ultimate goal.

Man come and go, but feeling self-worth and having good relations with their bodies is carried through the life. They confessed that the word ‘single’ does not scare them any more.

‘My body is my most long-term partner.’

'My husband might leave me somewhere in future, I think. Say that something is wrong with me. But I am confident, and I am happy that I didn't lose myself in him.'

'I used to be very unconfident about my appearance, and tried to adjust myself to the tastes of men I dated. I dated a fitness trainer and I hit my record 51 kilos. But I was angry and miserable and could not sleep. I worked out a lot, drank energy drinks and took weight loss pills to do that. I looked at myself with my man's eyes and saw only flaws. But when I became single, I realized that everything is amazing.'

'Pamela Anderson was their sexual ideal, and I even wanted to have breast implants. Now I think that it is not the outside that matters, and men fall for who I am as a person. It is interesting to hang out with me.'

Main values

Although I interviewed women from different countries and marital status, I found out that they share the same values and attitudes on body confidence.

Need for belonging and positive comparison

Women admitted that they value the feeling of togetherness and being part of a group of like-minded people.

'It is great when you have a community where you don't compare yourself with others because everyone is the same, and no one is looking at you with curiosity.'

'I felt my best body-wise when I was in a state rowing team, and everyone around me was as tall as me.'

'I try to look for people like me. Sometimes I ask my friends: 'Does this girl look like me? I am not that bad.'

Accepting and working with what they have

Over the years women realized that they might not ever lose weight or fix some features that they consider ugly because that's how their bodies are, and it is better to bring out best sides.

‘I learnt to accept my body. Maybe it is a Stockholm syndrome.’

‘What is a dream body? You are given a certain body and you need to learn what to do with it.’

‘If you are short, curvy and dark-haired, you will never look like young Kate Moss.’

Shopping as a body dissatisfaction factor

During the interviews, women admitted that they feel worse when they go shopping. And this is caused by a variety of reasons:

- **brands do not provide sizes and fits considering individual parameters**

Individual parameters vary greatly. The problems women described were:

•clothes don't fit in the chest or hips,

•if clothes fit in the hips, they can be too loose in the waist,

•clothes are too short or too long,

•clothes of bigger sizes are very often made just by scaling up a small size and don’t consider that a woman wearing XL can have a larger chest.

’25 jeans size is the smallest, but I look like I have pooped in them.’

‘Clothes are made for bad posture. I look like I am wearing a pillow cover. Yes, I fill in the clothes space, but it looks so unnatural on me.’

‘There are so many things that don't look good on big chest. And no clothes can make me look as if I have a normal figure.’

- **finding sizes takes time and lots of energy.**

Women admitted that they need to try hard and try on a lot of clothes before they find something that fits. For them, it is a trial and error process, and errors are more common. And for many of them, who fall out of the standard size grid it can result in having nothing to wear or having no particular clothing items, for example, swimming suits for small chest and hips, jeans for tall women.

‘I have the same model of underwear in different colours because it is impossible to find anything else. I found it accidentally. I saw that they were much smaller than the size written on the label.’

‘When I go shopping I expect that most likely I won’t find anything. I take all the clothes I like, and with this pile I go to the dressing room, hoping I will be lucky.’

‘Once I found mom jeans that fitted me, but I had no money at that time. I didn't buy them and the brand stopped making them. And it took me half a year to find a proper pair.’

- **shopping causes lots of negative emotions**

‘I think it is humiliating to look for XXS clothes. It is humiliating when you realize that you can dress yourself in kids department only.’

‘Sometimes I feel anger, pain or disappointment.’

‘Once I had a very fancy party and needed a cocktail dress. I cried in a dressing room because I could not find anything.’

- **it is not possible to evaluate how clothes will fit without trying them on**

Online shops demonstrate clothes on standard models, even if they are plus size or tall. And based on them it is still hard to imagine what clothes would look like on a particular woman’s body. Women admitted that they send lots of clothes back and don’t shop online because of frequent returns.

‘Sometimes I order delivery, and realize that a dress on a model can not look the same way on me.’

‘Now I can see clothes on models of my size, but not my height. Maybe on the website dwarf.com'

- **finding sizes require knowledge many women don’t have or have no time to acquire**

It takes time to learn about brand's sizes and characteristics, train your eye to see fashionable clothes and learn to pair them together.

'I know certain brands that will definitely fit me. For example, Levi's have size 24 jeans. Even 23.'

'I love petite clothes line. You just try them on, and if something is supposed to be knee length, it will definitely be knee-length. I even don't need to see it on a model.'

'New look has 6 size, many other brands start from 8.'

'I don't know how clothes should fit and take a friend with me.'

'Once I impulsively bought leather leggings, and then could not make up any look. A friend of mine suggested wearing them with a white shirt, and it has been one of my favourite looks recently.'

•clothes influence professional life

Failing to find clothes results in failing to create a personal style, which can influence professional and social spheres of women's lives or what message they convey through clothes.

'People do judge by appearance, especially at work. And if I wear a T-shirt and leggings (because I can always find them in my size), I won't be seen as a professional.'

'I think I express myself more through accessories. Because It is hard to do the same with clothes.'

'I am especially unhappy with how I look when seasons change. You have just got used to winter and then spring comes. Everyone looks beautiful but me.'

'Am a grown woman, I have money, but I still dress like a student.'

5. Desk Research

Interviews provided me with sufficient amount of evidence that shopping for clothes is an everyday trigger for many women, and it goes far beyond wasting time and not finding the right size, but has a negative impact on emotional life. But my research would be incomplete without seeing a bigger picture behind collected insights. To see how the problem is shaped by psychology, culture and retail industry I conducted desk research and set the following aims:

- explore what industry factors make shopping for clothes complex and not capable of catering for all women,
- review what characteristics of the retail industry have an opportunity to be adjusted by my solution,
- discover why failed shopping experience creates so many negative emotions from the point of view of culture and psychology
- have an overview of existing solutions
- see how women challenge the situation and what solutions they come up with

5.1 Retail industry overview

Fast change of collections

Zara offers 24 new clothing collections each year; H&M offers 12 to 16 and refreshes them weekly. Among all European apparel companies, the average number of clothing collections has more than doubled, from two a year in 2000 to about five a year in 2011 [10]. This is aimed at 'creating the feeling that anything the customer could see at any given moment as an article they might possibly buy would cease to be available within a few days. [11]'

Usage of psychological mechanisms

Apart from scarcity, shops rely on coping mechanism called "**compensatory consumption.**" This mechanism means that even when a woman can only fit larger sizes at a specific retailer, she'll purchase other products, such as jewellery or

make-up to boost her self-esteem. They also exploit ''**self-discrepancy gap**' - a difference between the way you see yourself and the way you'd like to be. As April Lane Benson, a psychologist who specializes in the treatment of compulsive buying disorders, put it: 'it nudges us to the world of fantasy, a world where what we wish to happen takes precedence over what we can reasonably expect to happen. Who you are becomes confused with what you have' [12]. Those two phenomena manifest themselves in shopping as a way to camouflage a negative body image and strive to look like somebody else or make yourself a better version, instead of accepting who you are.

Vanity Sizing

Today, clothing manufacturers are often using "vanity sizing," the labelling of clothes with sizes smaller than the actual cut of the items. So, those size 34 slacks might really be closer to size 36, and perhaps even bigger [13]. (Figure 6). It is done both by mass market and luxury brands to increase sales and is based on shoppers' desire to feel slim. Vanity sizing has a negative side, resulting in smaller women being removed from standard size range and introduction of 000 size of clothes and setting an even more unattainable body ideal to those having body dissatisfaction.



Figure 6: Vanity size in action: size 12 in different Credit: @chloemmx

Size as a brand identity

Brand's clothing shape, proportions of each size and eases convey a brand's philosophy and help a brand to stand out from other brands. In other words, brands sell not only clothes but also a certain lifestyle and a certain fashion body.

Absence of Standards

Throughout history, there were attempts to create a universal size charts. For example, the US National Bureau of Standards published "Body Measurements for the Sizing of Women's Patterns and Apparel" in 1958 based on the 15,000 women interviews. Those standards were updated in 1970 and were ditched in 1983, after being proven not to be realistic. The reason why any standardization fails hides in our genetic variety of sizes and proportions (Figure 7). As a result, modern brands have to determine their size charts themselves.



Figure 7: Body diversity by We Are Living Cute's

5.2 Psychological and socio-cultural factors of shopping and fashion

Being different while being the same

According to Mari Grinde Arntzen [14], fashion is run by two human needs: stand out from the crowd and at the same time, comply and be like others. On the one hand, we want to demonstrate our personality and our individual features, but on the other hand, we want to be a part of society or a group of people. And every day we are balancing between those two needs, and this balance is powered by fashion.

Fashion as communication

Clothes act as a means of self-identification, because they are a way people interact with the world, transmit information about themselves and let other people interpret it. Nowadays, due to the overload of information and lack of time, people are guided by instincts more and thus tend to judge people more by their appearance. Otherwise, people would not be able to manage in this world.

Fashion promotes comparison

'Considering that the majority of social media and advertisement is image-based, it becomes apparent that social comparison could be a serious issue for its users' mental health and well being'. [15], which results in negative changes in self-esteem, body satisfaction and mental health.

Beauty as social must

Naked body by itself does not bear any valuable information. Clothes serve as a body that we want to show to this world, hide our flaws and display a self-made and constructed ideal. This concept is especially evident in the fact that the fashion system itself is aimed mostly at women. According to a theory called 'Beauty system' [16], women are considered beautiful and feminine based on the amount of effort they put in themselves. That's why women indulge themselves into beauty practices and buy more: naturally looking and unconstructed female appearance is not considered beautiful. But because it is a false identity imposed on women, it creates a conflict between what an individual wants to be and how others see her, and results in experiencing shame.

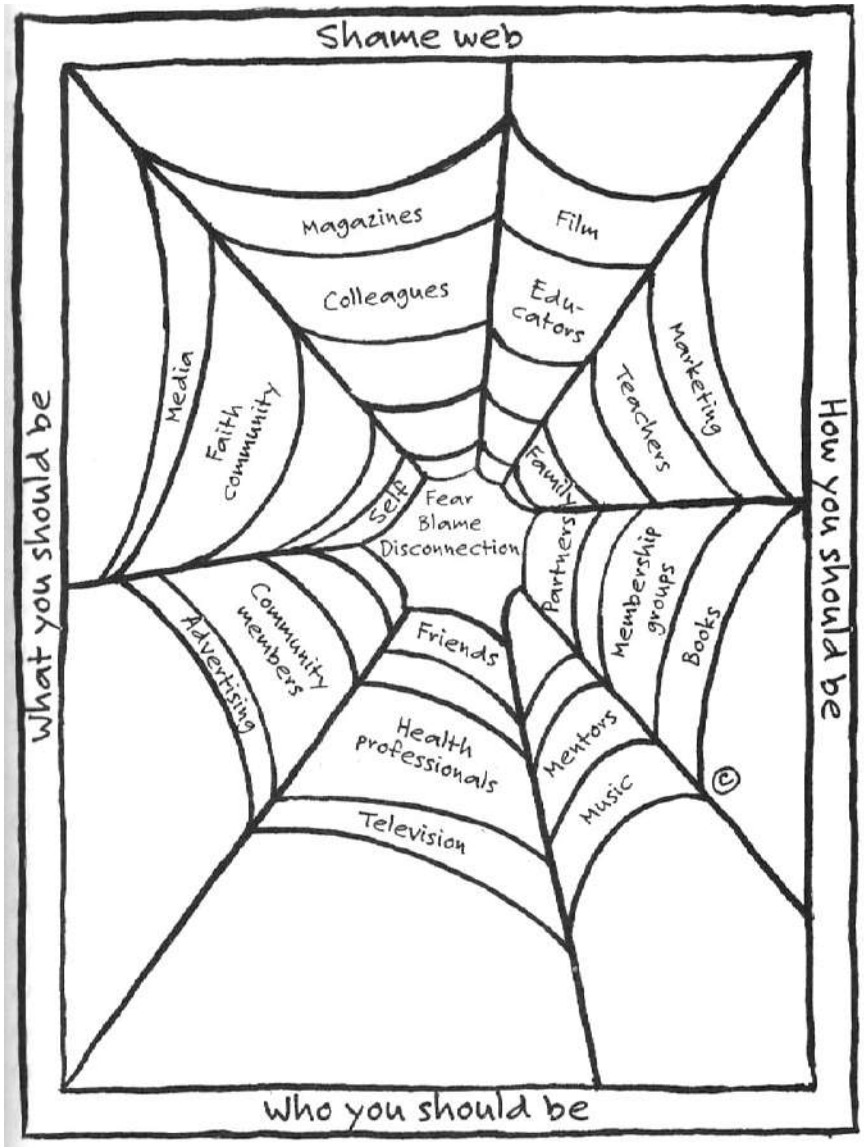


Figure 8: Shame web

Shame

Dr. Brené Brown, a research professor who has spent the past two decades studying courage, vulnerability, shame, and empathy, states that modern society does not talk about shame, it is a 'silent epidemic' [17].

She describes it in a number of **definitions**:

- A feeling that you can not talk about because then everyone would know your dirty little secret.
- Shame is being rejected
- You work hard to show the world what it wants to see, but your mask falls and everyone sees unlikeable parts of you.
- Shame is feeling like an outsider - not belonging.
- Self-loathing
- Shame is like a prison, you deserve being in because something is wrong with you.
- Shame is being exposed - the flawed parts of yourself that you want to hide.

In other words, shame is an extremely painful feeling or experience of believing we are flawed and as a result,

unworthy of belonging and acceptance. It is an emotion of self-evaluation, and it shifts self-reflective thoughts from 'I did something bad' to 'I am bad', putting a focus on who we are, and if we repeat saying it to ourselves, we start believing in it and indulging in self-destructive behaviours. It creates feelings of fear, blame and disconnection (Figure 8).

According to Brene Brown (Figure 9), women need to develop **shame resilience** by:

- recognizing shame and understanding triggers,
- practicing critical awareness,
- reaching out, for example build connection, mutual support, shared experiences, acceptance and belonging, for example expanding social network or creating support groups, understanding cultural, political and economic influences on our personal experiences,
- speaking shame.

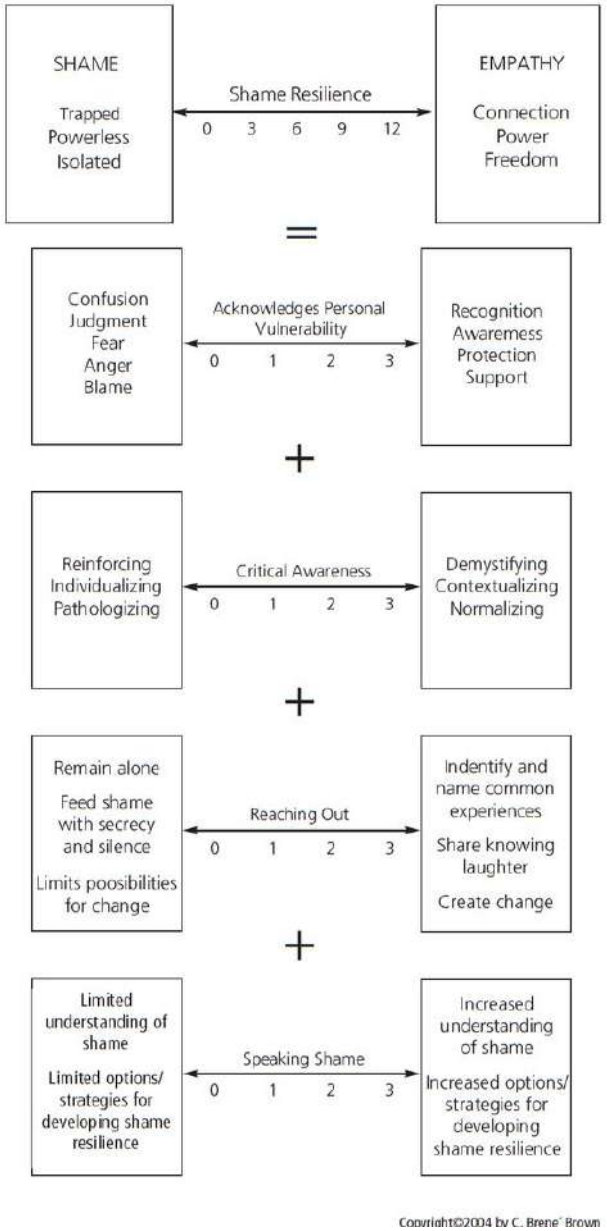


Figure 9: Steps to shame resilience

5.3 Solutions initiated by people

Women themselves try to solve sizing issues with means available to them.

Models, bloggers and body positivity activists try to raise awareness of this problem and share their personal stories of failing to find clothes that fit. For example, plus-size model Marielle Elizabeth (Figure 10) wrote her story in the article 'Apparently, I'm Too Fat to Ski'.

'I've always wondered why "bikini season" is the only time of year synonymous with size-based self-loathing. Seriously, have you tried being fat in winter? I speak as an expert: I know the true desperation that comes with trying to find plus-size snow pants. I have wept in a sporting goods store, with the temperatures dropping outside and a romantic getaway that required a ski jacket on the horizon. I have optimistically moved a pair of snowshoes from three apartments, despite not owning the clothes to wear with them for over a decade. Oh, I am bitterly aware that winter comes with a size limit.' [18].

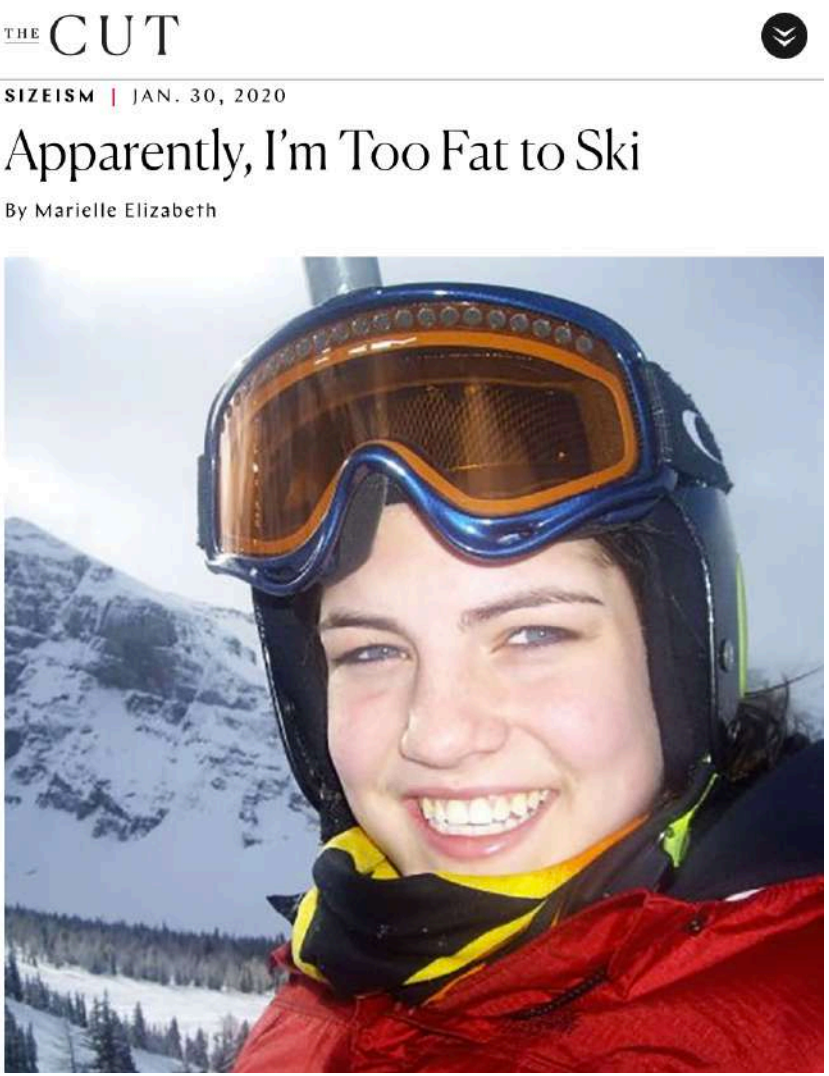


Figure 10: Marielle Elizabeth and her story

Katie Sturino (Figure 11) and other influencers raise size problem on Instagram and review brands sizes, show that fashion has no size and anyone can look fashionable.



Figure 11: Sturino's posts on the problem of sizes, taken from her instagram @katiesturino

Such posts create emotional responses and become a place where women share their struggles and life hacks to be seen by other women (Figure 12).

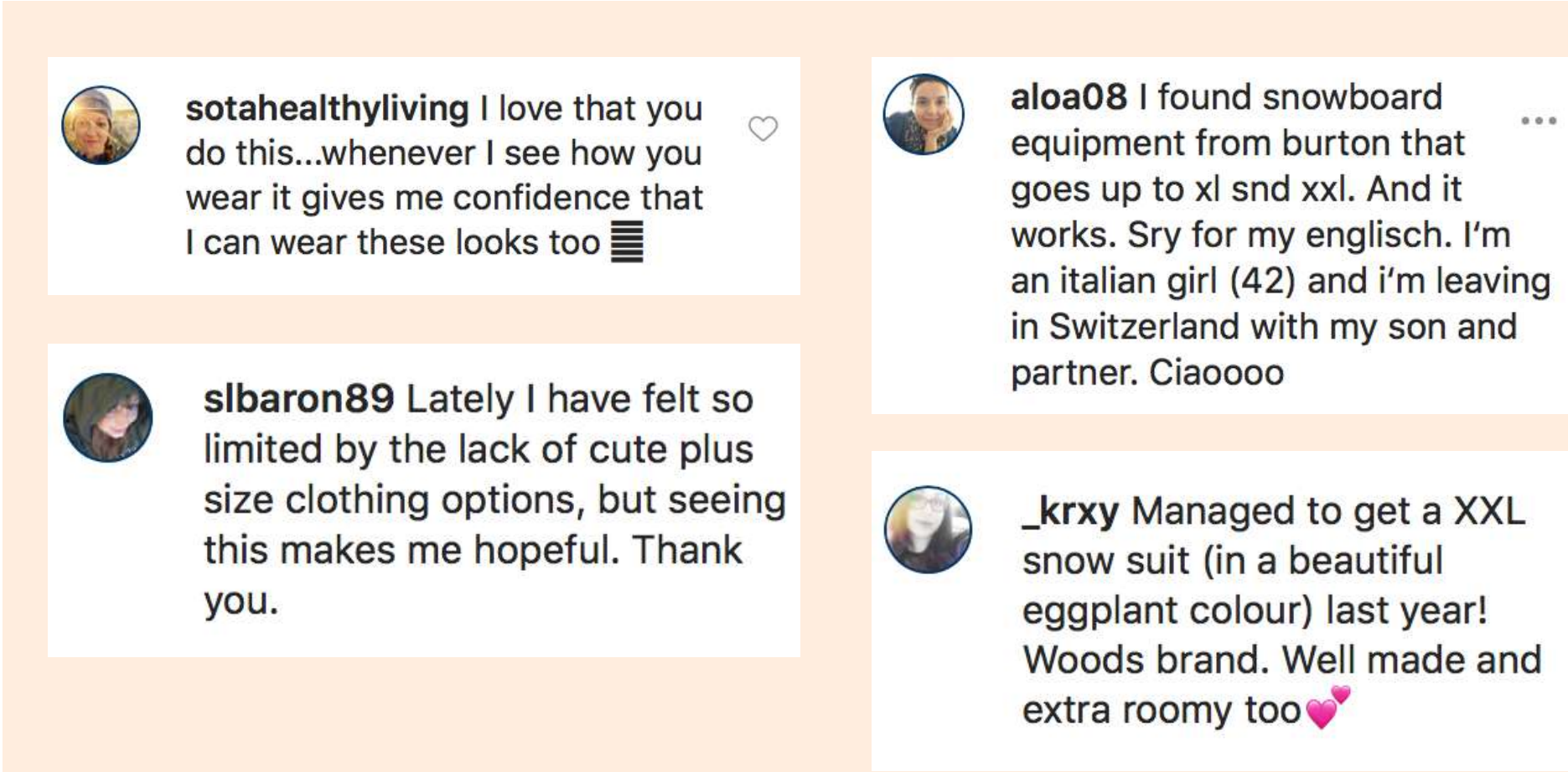


Figure 12: Examples of comments under Sturino's posts

Women also create specialized groups, where they share life hacks and findings. Below is an example of such a group (Figure 13), called ‘Not your size’. The title of the groups is: ‘This is a group for fat women or women who think they are fat. Or for those feeling ‘like something is wrong with them.

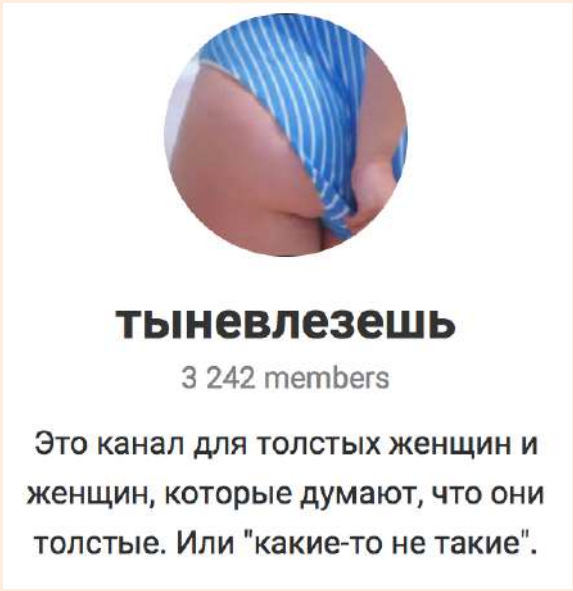


Figure 13: Example of specialized groups

5.4 Overview of technological solutions

I reviewed technological solutions available in retail and offered to customers and found out that they all fall into 3 categories: AR-based solutions, AI-based recommendation systems and tangible solutions.

Augmented reality

Fashion brands use AR technology in a variety of ways:

- showing products in real life via real models to help customers evaluate the fit or see how an actual product could look outdoors or in real life scenario (Figure 14),
- installing smart mirrors in stores to provide visitors with dressing room alternative, offering clothes combinations and providing instant sharing. (Figure 15),



Figure 14: Asos Catwalk AR



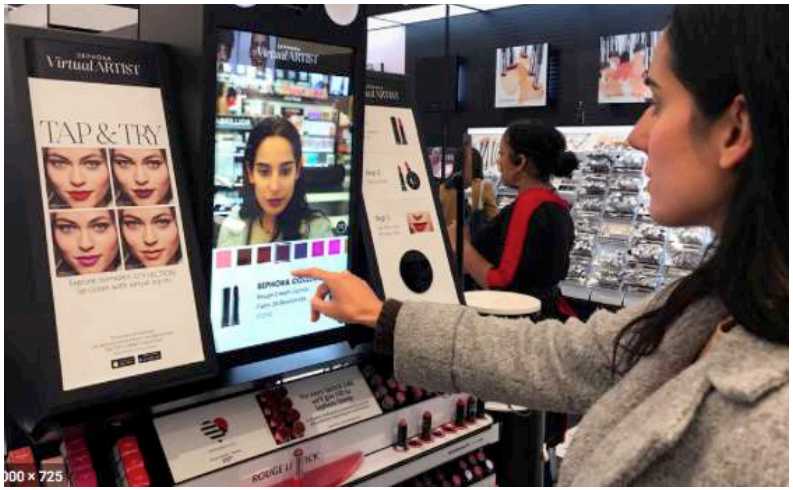


Figure 15: Smart mirrors

- bringing dressing rooms into homes of their customers. It is done by either creating a mannequin or a virtual avatar based on a customer's height, weight and body measurements, or by helping to visualize clothes, shoes or jewellery on a body through a smartphone camera. Examples of such solutions are Gap fitting room(Figure 16) and Wanna kicks app (Figure 17).

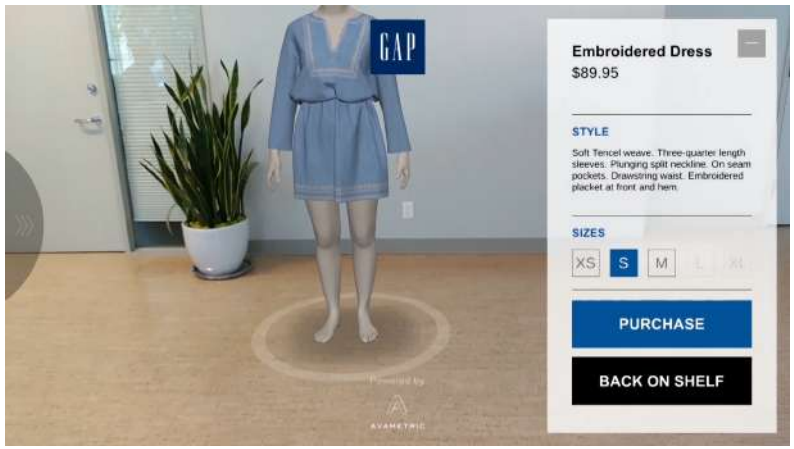


Figure 16: GAP fitting room

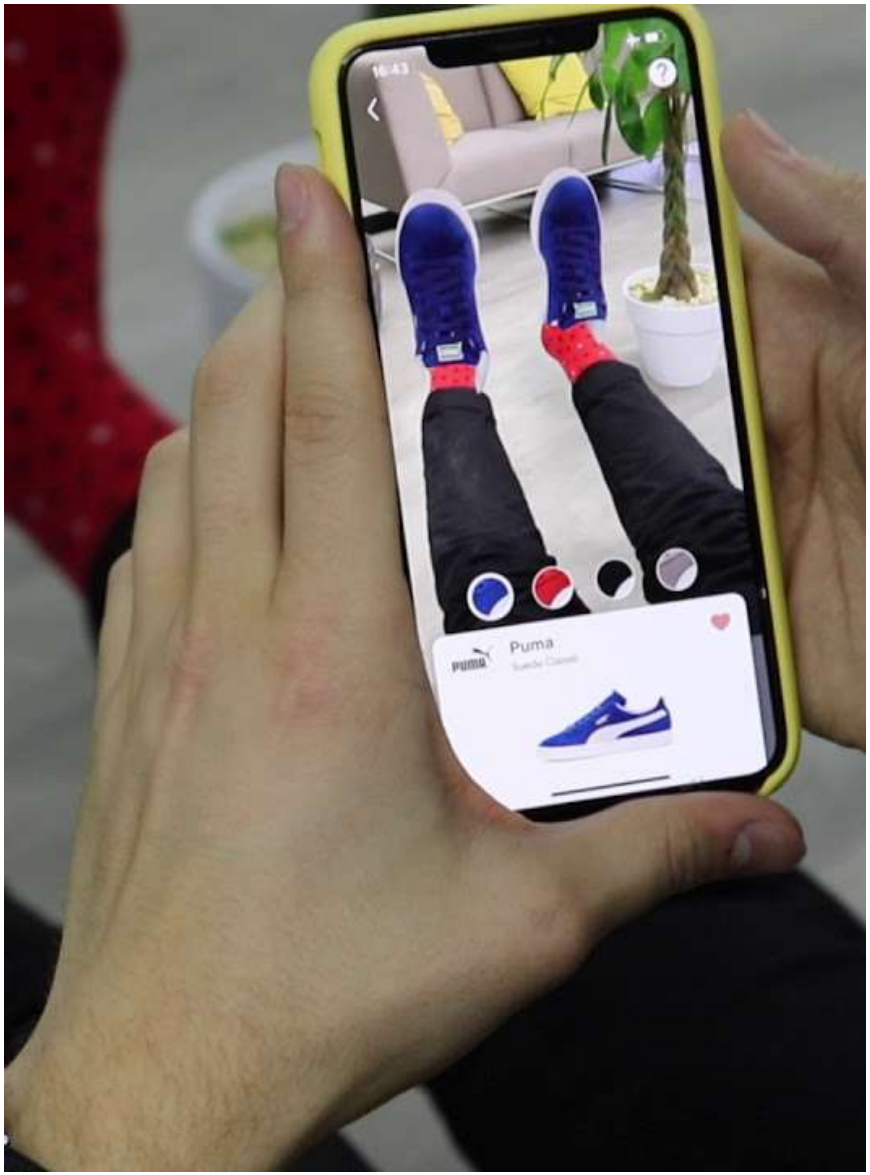


Figure 17: Wanna kicks app

Although AR visualization can help customers to see how clothes can look on people, it can not be entirely accurate and does not take into account materials, shape and surrounding light, because developers either use images recognition and the body is seen as flat by the algorithm, or create an approximated 3D model of a human body (Figure 18).

Even ASOS' 'See My Fit' Tool (Figure 19), which was launched in 2020,lets users see clothes on models of 16 different body shapes using AR, but the image is not completely accurate.



Figure 18: Drawbacks of AR in placement of objects by Design eLearning Tutorials



Figure 19: ASOS 'See my fit'

Tangible solutions

Estonian start-up Fits.me uses a mannequin (Figure 20) that shifts into 2,000 different body shapes to let users try clothes online. Each size is placed on the robot and viewed on all the body shapes. Pictures taken during the process are stored in a database and when customers shop online and add their measurements, they can see corresponding pictures.

Artificial Intelligence solutions

Artificial Intelligence solutions are mostly focused on online style and fit advise. For example, Intelistyle uses an algorithm to recommend styles and outfits based on preferences, eye colour, skin and body shape online. EyeFitU app uses same approach, but also caters for in-store customers: users can look up an item in the app and find the size they need. Outfittery uses questionnaires and a combination of AI and real-life stylists to personalize shopping. Canadian/German start-up Twenty Billion Neurons (TwentyBN) created an avatar (Figure 21) that can interact with shoppers and advise on stock and pairing.



Figure 20: Fits.me solution



Figure 21: Millie

5.5 Desk research summary

Interviews followed by a desk research helped me see a bigger picture behind an everyday shopping experience. I found out that:

- from the point of view of **technology,** solutions reached a high level of visualization and accuracy, but still are unable to offer realistic recommendations/size predictions. They are run on retailer's initiative and aimed mostly at online shopping,
- from **retail** perspective, it is still impossible to adjust ready-to-wear clothes to all figures and sizes and change approach to sales in the near future: it is not only a complex economic and logistics system, but also a system exploiting psychology to drive sales.
- on the level of **psychology,** not finding suitable clothes is an emotionally painful experience for women, because it means failing to conform to social norms and convey the right message about themselves. Taking a bigger size or not finding anything to wear starts the process of self-comparison with media images and other people, resulting in feeling of unworthiness, shame and being alone with those kinds of problems.
- from **people's** approach, creating groups and leaving comments is the most common way how to share information on clothes, and an unconscious way to practice shame resilience in a form of accepting vulnerability, practicing empathy, sharing personal stories and building connection. The biggest drawback of this solution is that findings are location and time based, and they lose their relevancy very fast. Also they require preparation for shopping: looking through the tips in advance and memorizing/saving useful information to use when going shopping.

6. Ideation

6.1 Design principles

Design principles are 'the guardrails of your solution—quick, memorable recipes that will help keep further iterations consistent. These principles describe the most important elements of solution and give integrity and form to what you're designing' [19]. I formulated them based on my research.

Usage of the current paradigm

- My solution **should not** be aimed at redesign of retail system or its parts. It means that I can not redesign sizes creation because it can become one more means of psychologically nudging people into buying and lead to the creation of even more insecurities in women
- My solution **can** take advantage of shops organization, for example, how clothes are displayed or how often they change collections.

Promotion of positive comparison

- My solution should aim at normalization of female bodies, showing real people with real bodies in real situations and support values common to my target group, described in user interviews results.

Building a sense of community

- The solution should incorporate principles of shame resilience, formulated by Brene Brown, such as practicing mutual support, sharing experiences, this way creating a feeling of everyone 'being in the same boat'.

Real-time contextual information

- Solution should provide relevant information during shopping.

Non Intrusiveness

- Solution should not eliminate the pleasurable parts of shopping experience, but rather be an assistant, usage of which is initiated by users.

6.2 Generating opportunities

According to IDEO ‘we use the How Might We format because it suggests that a solution is possible and because they offer you the chance to answer them in a variety of ways. [20]’ I translated my insights into the following opportunities:.

- How might we help women find the right fits and sizes?
- How might we help women cope with negative emotions while shopping?
- How might we build positive comparison while shopping?

6.3 Persona

A persona is an ‘archetype, not a real person, but a synthesis of facts and observations about real users that leads to a memorable character. By creating these characters and taking them through scenarios, you and your stakeholders can gain a sense of familiarity and empathy with your users. [21]’ The persona I am designing for is based on the interviews and desk research results and comprises characteristics and stories common to all women I interviewed.

Meet Kate

Kate is 29 years old, works in an HR department in a company (Figure 22).

She is short and wears size XS or XSS, depending on the brand. Most of the brands overlook her size or make clothes bigger than stated, and she needs to be very inventive and spend lots of time to find something. She can not shop in kids departments because she has a dress code at work and needs to look professional. Ordering online is not an option for her either because it is hard to evaluate what she will get based on the picture and size description, and very often she sends all the orders back.

When she goes shopping, she takes everything that might fit, and with this big pile goes to the dressing room. While trying things on she gets very disappointed and sad, realizing that everything is too big or too long. She looks at other women, walking out of the dressing cabin to show the outfits to their friends or waiting near the counter, and thinks that this is unfair and that everyone is normal but her.



'I am invisible for the current size chart.'

'I wore only leggings for half a year, because I couldn't find any trousers or jeans.'

'I have no swimming suits.'

'One dress out of 15 in Zara can be my size'

Figure 22: Persona

She is an active social media user: has an Instagram account, likes to post pictures and stories, tags friends in them.

She likes new technologies, and bought a drone to make videos when she made a trip around Asia together with her partner last year.

She believes that sharing is caring. When they went to China, lots of people helped them without asking anything in return, even though they didn't have to and could simply ignore them. That's why she hosts couch serfers, believing that it is a way to pay back to the universe and people.

Her goals are:

- Feel good about herself when going shopping.
- Spend little time on shopping and devote spare time to activities she enjoys instead.

Her frustrations are:

- Does not know how clothes should look on her and where to find them.

- Spends too much time shopping and often ends up buying nothing.
- Has no idea what people of her height and body shape should wear.
- Gets very dissatisfied about her body after shopping.

6.4 Brainstorming

'A brainstorm is a distinct segment of time when you amp up the generative part of your brain and turn down the evaluative part. The intention is to leverage the collective thinking of the group' [22]. Brainstorming can be carried out using different techniques, such as brainwriting, "five whys", negative brainstorming and etc. For my project I used 'Yes, And!' technique to have more structure and constrains within the session and be able to build on top of others people's ideas.

I invited my coursemates and shared my design principles, How Might We Questions and persona with them so that they had a better context for brainstorming and build empathy on (Figure 23). I was especially interested in a way men would approach the problem and see how much they can empathize with my persona, which would be helpful preparing my thesis presentation.

For each of my How Might We Questions my coursemates wrote down and idea, then we in turns started adding ideas to already existing ones (Figure 24). After the ideation I reviewed all the ideas, and grouped themes arising from them:

- using a helper: AI assistant or a friend
- showing clothes by light or voice interface
- using a virtual avatar in the shop



Figure 23: Brainstorming session

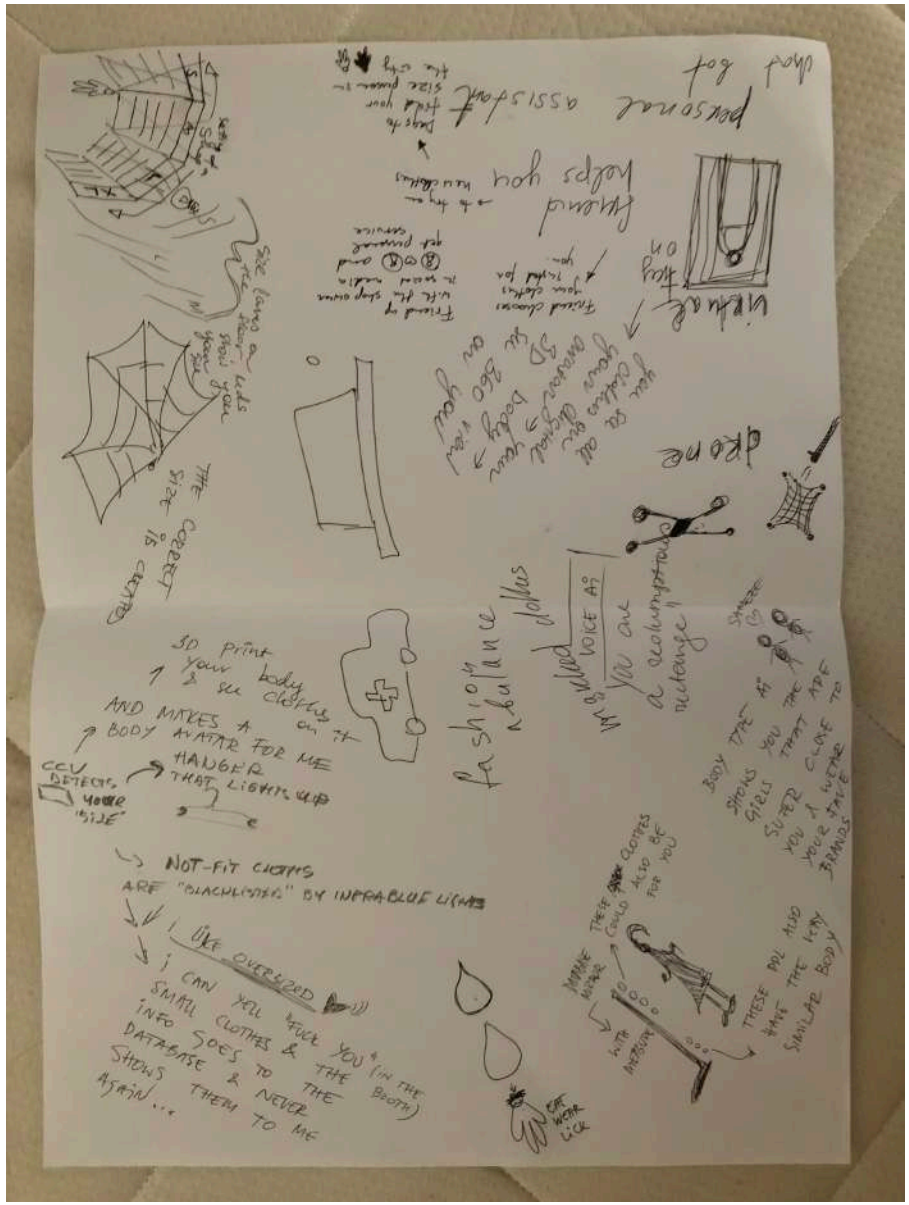


Figure 24: Brainstorming results

6.5 Ideation through sketching

Brainstorming with other people showed that my design principles, especially not changing anything in the current retail system, are constraints that are easy to breach. And most of ideas breached at least one, for example suggesting to change the delivery method or reinventing how sizes are defined. In addition, brainstorming with coursemates showed that effective ideation required a deeper involvement in the project and knowledge of the topic. That's why I decided to ideate on my own using sketching. 'The process of sketching is a kind of cognitive scaffolding, a rich and efficient way to off-load part of the cognition, especially the mental visualization, to physical artifacts in the world. A sketch is not just a way to represent your thinking; the act of making the sketch is part of the thinking.' [23].

First concept

First idea that seemed relevant to me was that a woman has a body profile in her phone and when she goes into the shop, she scans a size shown on the label and the app checks the parameters, compares them with her measurements and says if clothes will fit or not (Figure 25).

I decided to validate this idea by using ASOS style assistant and size guides available online. I selected trousers on Asos similar to the ones I already have and used their size assistant. Unfortunately, it didn't manage to help me, though the system had my measurements and some information about my body (Figure 26).

Then I looked at the size guide published by Collusion brand, measured myself and found out I need size UK18 - 20. But I have them in UK16 already and they are even a bit big.

This small experiment and industry overview done earlier proved, that no-one knows if clothes will fit or not unless tried on, and made my concept irrelevant.



Figure 25: First concept

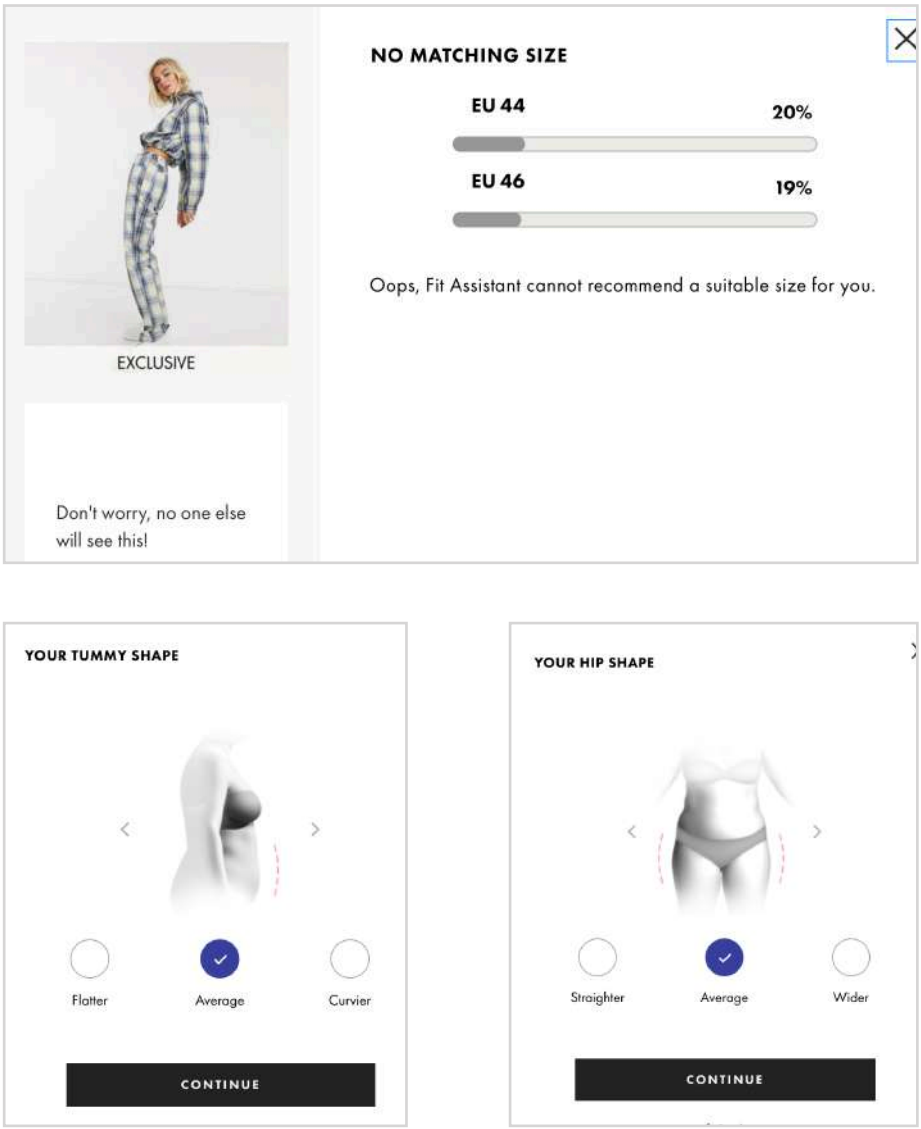


Figure 26: ASOS size assistant

Second concept

For the second idea (Figure 27) I got inspired by my Belorussian friend, living in Korea. Once she told me: *'When I see a fat Korean girl, and you know they are rare, I want to approach her and ask where she found her clothes.'*

My idea was that you could see what women are wearing, what sizes and what brands. But when I drew it, I started thinking that it is a bit 'stalkish', information may not be up-to-date, as the collection is long gone. What if a person passes-by quickly? You won't have time to check your app? Will information about passers by be saved constantly on the phone? I started having lots of questions, and by questioning my concept, realized that it is irrelevant.



Figure 27: Second concept

6.6 Final Concept and Validation

After those first trials I started thinking of possible ways how to create an individual informational overlay, that could be accessible only by women having something in common, this way being able to relate to each other. And I came up with an idea that AR could be such a perfect overlay, and each shop - a place for women experiencing same size problems.

Kate would go to the shop, and through the app find what other women wearing XS or XSS size tried on. She could see all positive and negative reviews left by her same-size community. In AR mode she can just look around in the shop and see what is there (Figure 28).

To validate my idea as early as possible, I decided to show my sketches (Figure 29) to my target group (Figure 30).



Figure 28: Concept



Figure 29: Sketches shown to women

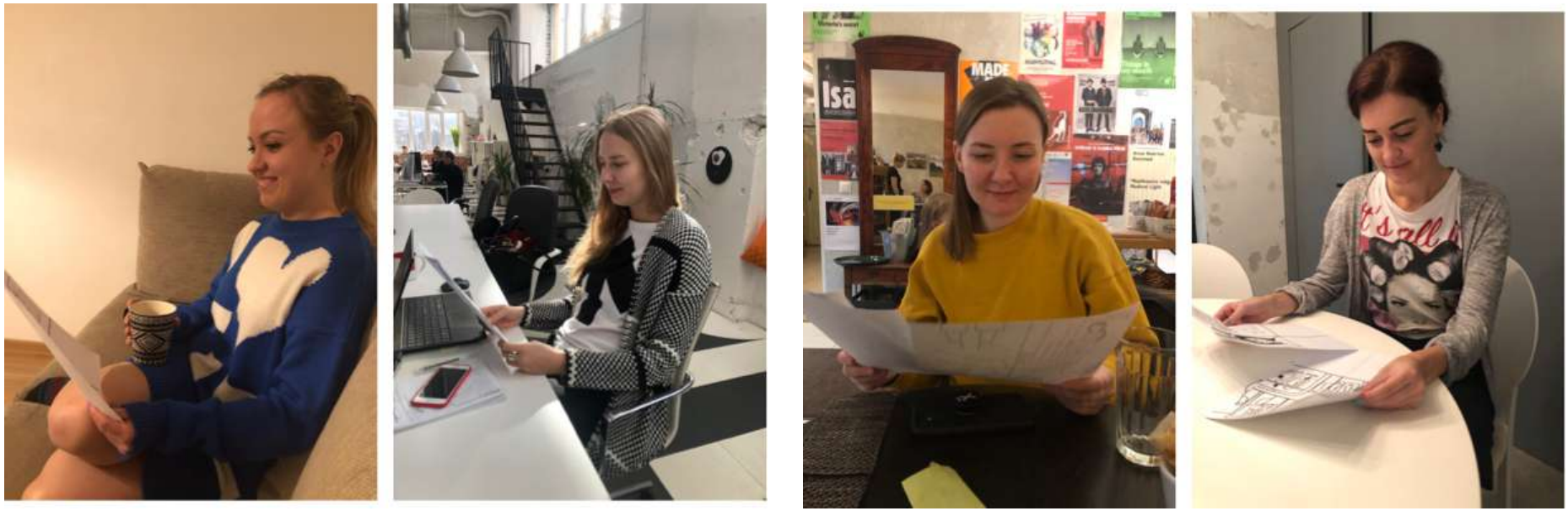


Figure 30: Participants

My aim of the validation was to:

- see how women relate to the situation I have sketched,
- find out what was their attitude to the idea,
- lean about device preferences.

Based on the discussions I grouped my findings into 4 categories (Figure 31).

I found important points that need to be considered when working on a solution:

- There needs to be a good on-boarding, explaining not only how to use an app, but also why the information is relevant to you and where it is received from
- Women found the idea a time saver and a way to reduce shopping dissatisfaction
- No one was interested in AR glasses solution, because at the moment they are expensive, and AR glasses launched previously failed, and women were skeptical about the new ones.

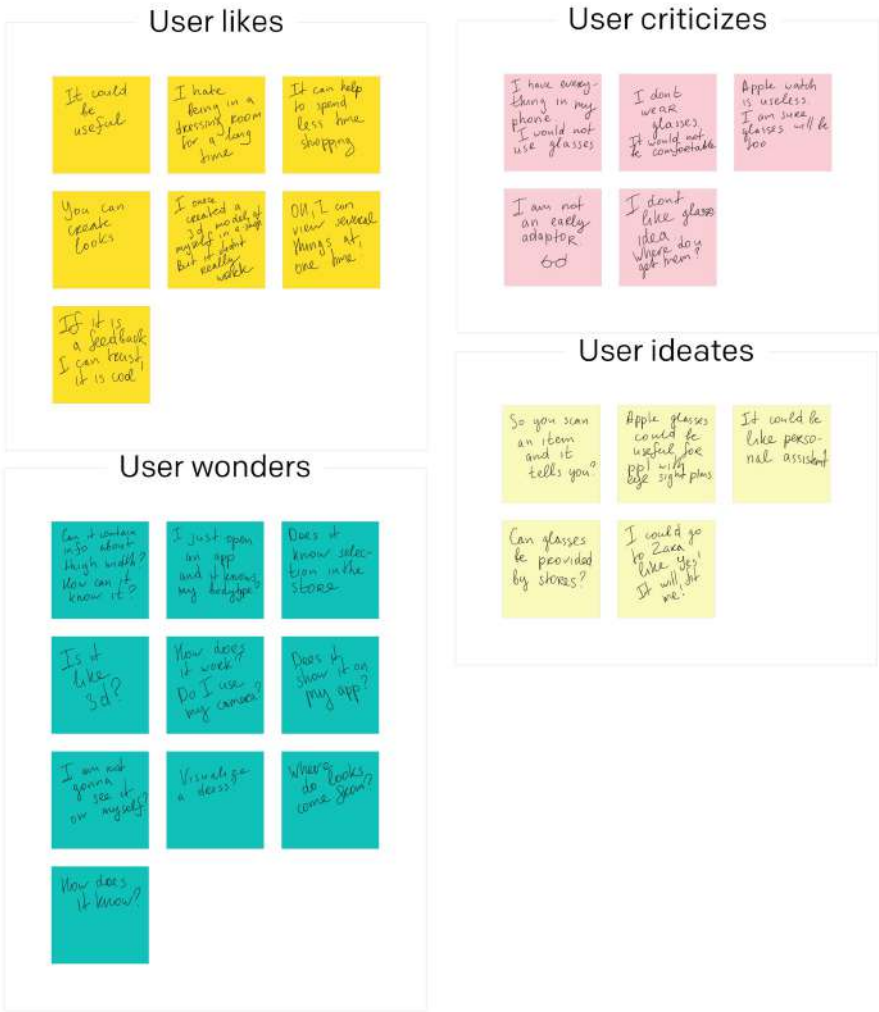


Figure 31: Feedback capture matrix

Seeing how women ideated was extremely valuable to me because it helped me to think of other variants of the concept. What if it was initiated by shopping malls and what could be the way? This way we can embrace a big number of shops. But while sketching (Figure 32) I realized that most of the ideas are either too complex or expensive, and as a result, not realistic.



Figure 32: Idea exploration based on discussion with target audience

6.8 Conclusion

The process of ideation reminded me of a puzzle. I had all the information from my research and design principles, but the concept did not come together because one piece was missing. Most often the principle that was not covered by solution was using the current retail system and not applying any changes to its parts. Since I was well determined not to change or omit any of my design principles, I had to produce as many variations as I could through sketching, brainstorming and showing ideas to people. Thus I managed to see different combinations of my principles in action until I found the concept able to combine them all.

7. Concept development

7.1 State of Augmented Reality

After having created my concept, I needed to conduct additional desk research and find out how popular AR technology is and what are the technological capabilities of Augmented Reality and the nearest future trends..

Popularity of Augmented Reality

AR technology has become democratized after introduction of Snapchat's Lense Studio and Facebook's Spark AR Studio, allowing everyone without developer skills design, build and share AR experiences. It opened new opportunities for brands and their online presence and increased involvement and interest of users in Augmented Reality. For example, in the first year (2017) of Lens Studio, creators have submitted more than 250,000 Lenses that have been viewed more than 15 billion times [24].

Current state of technology and it affordance

Apple's ARKit and Google's ARCore software development kits (SDKs) were introduced in 2017 and have standardized the development tools, making mobile AR app creation accessible to the mass audience.

Latest Apple's ARKit 3 can [25]:

- naturally show AR content in front of or behind people using People Occlusion,
- track up to three faces at a time,
- support collaborative sessions,
- simultaneously use face and world tracking on the front and back cameras,
- capture motion,
- detect up to 100 images at a time,
- get an automatic estimate of the physical size of the images,
- recognize objects in complex environments
- use machine learning to detect planes in the environment faster.

Hardware

Smartphones have already become powerful devices, capable of supporting AR experience, and over upcoming years will acquire new features and sensors. For example, Time-of-flight sensor, measuring the time it takes for something (be it a laser, light, liquid, or gas particle) to travel a certain distance, will be able to calculate a detailed 3D map of the room and all of the objects in it.

Return of AR glasses

After the release of Google Glasses in 2013 and their later fail on the market, no one has attempted to introduce real AR wearables, but instead introduced limited functions to glasses. For example, Snap has the Spectacles 3 that allows integration of filters and recording of 3D video. Amazon created Echo Frames, that let users access Alexa on the go.

Glasses can become a big game-changer in the industry with Apple, planning to release an augmented-reality headset in 2022 and a pair of AR glasses by 2023, and aiming to replace the iPhone "in roughly a decade". Supposedly, the glasses will synchronize with an iPhone to display texts, emails, maps, and games over the user's field of vision. Facebook is also working on glasses, and plans to release them in 2023, 2024 or 2025 latest.

Key finding

- AR technology is affordable and understandable by users.
- It can support all the functions needed for my solution.
- It can fully provide individual overlay to the current retail paradigm.
- Over the time my solution can evolve with the appearance of AR-enabled glasses.

7.2 Mental Model

I decided to use Indi Young Mental Model to understand how my solution can or can not support people and help to define my solution's functions. According to Indi Young, mental models can be applied to many different situations and stages of design and development. 'Say you already have prototypes and usability test results. Say the users just don't

get it. Sketching out a mental model will help you see exactly where your design veered off in a direction different than users.

What if your beta application is faring well and you want to know in which direction to move next? It's perfect timing to align functionality to a mental model and prioritize the gaps. [26]'

To create my Mental Model diagram (Figure 33), I went through the interviews that I made previously, and wrote down all the tasks I found.

In this approach "tasks" mean actions, thoughts, feelings, philosophies, and motivations that come up when a person accomplishes something or achieves a certain state. Then I grouped them and merged the ones that mean the same things.



Figure 33: Mental model

Under the towers I added functions (Figure 34) that can be implemented in my solution to help women carry out their tasks or spare them from unnecessary ones.

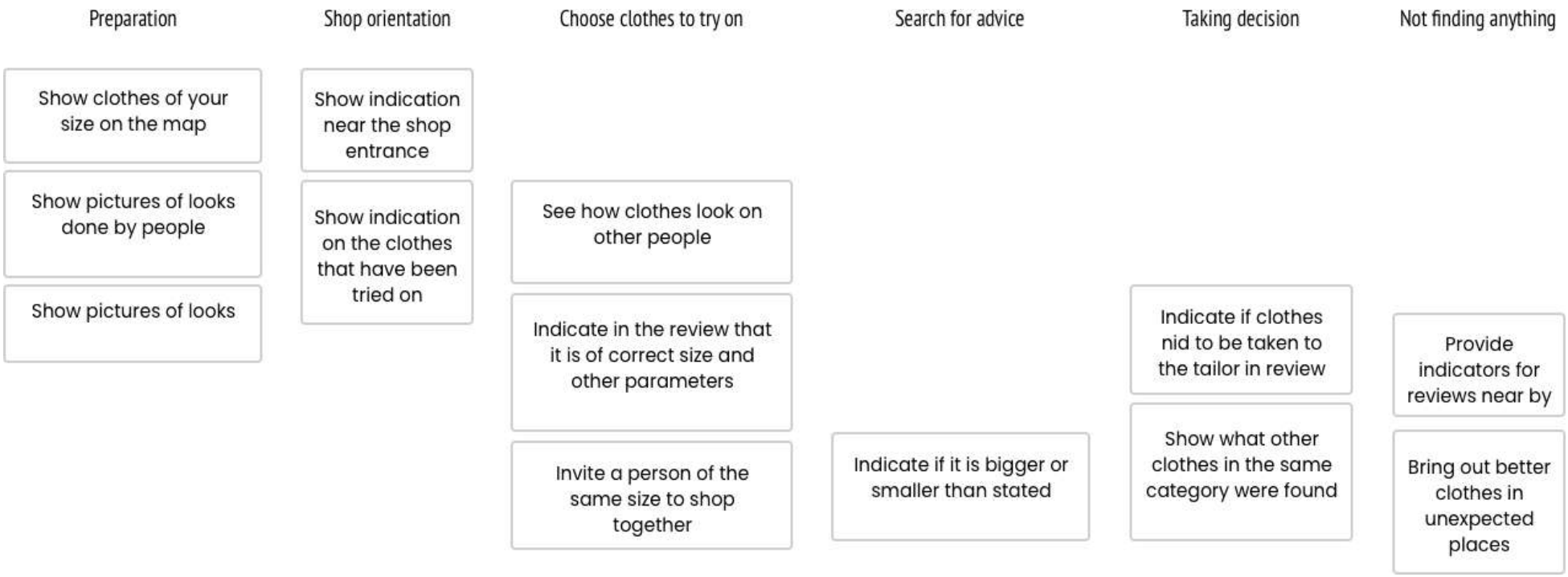


Figure 34: Functions covering users' tasks

7.3 Prototyping and testing

I decided to do 3 rounds of user tests with 5 users per round. According to Jakob Nielsen, 'you want to run multiple tests because the real goal of usability engineering is to improve the design and not just to document its weaknesses. After the first study with five participants has found 85% of the usability problems, you will want to fix these problems in a redesign. The second study with 5 users will discover most of the remaining 15% of the original usability problems that were not found in the first round of testing. (There will still be 2% of the original problems left — they will have to wait until the third study to be identified.[27]'

I decided to conduct usability tests online using Lookback (Figure 35). I found my test participants through a Russian Telegram channel 'Looking for brands' which caters for a group of women similar to my target audience.

Prototyping and testing - Round 1

For the first round of tests I created a paper sketch (Figure 36) of my concept and created a prototype using Invision to share with participants. I decided to leave it very open to interpretation because it is the last step where I can see user ideation, not being limited by interface functions. Then I formulated task scenarios I wanted to test:

- You are in a shop, and you want to find out if someone has already tried the dress you like and what feedback she left
- You tried on a dress, and it looked terrible on you. You want to leave a review.

After each session I had a follow-up interview to find out what women liked/disliked about the idea, and learn more about their shopping habits or problems that they face.

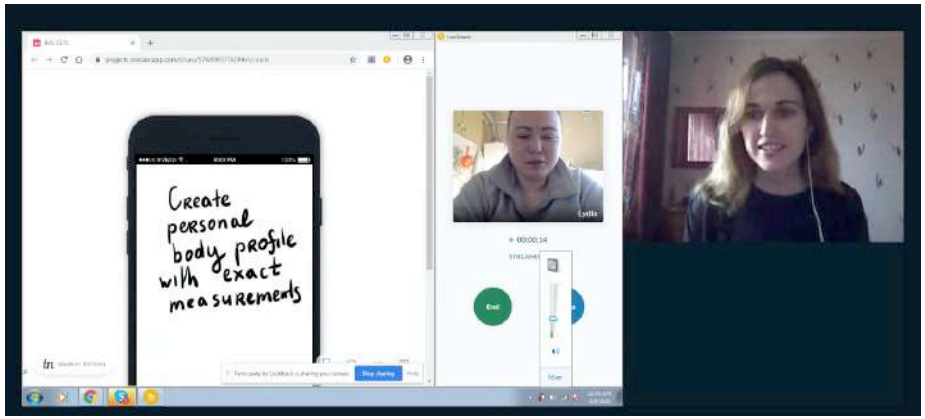


Figure 35: Lookback testing session

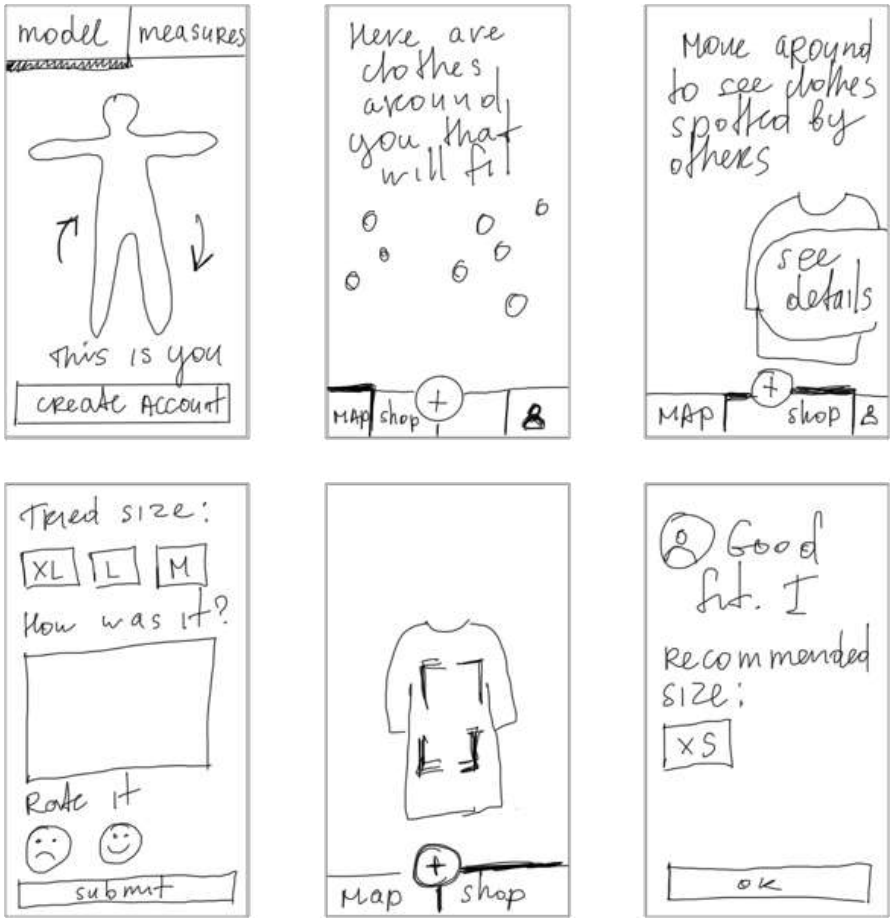


Figure 36: Paper prototype main screens

Results

Generally, women were interested in my idea and did understand the concept, but my paper prototype could not show the context of augmented reality well.

The recurrent issues were:

- all 5 women found copywriting confusing: everyone thought that learn more button was about reading information about the product
- 3 did not find how to add clothes review
- 3 found rating icons misleading and considered them to be a mood selection or too general
- 2 clicked on clothes on the map, although my idea was that they are not clickable.

Prototyping and testing - round 2

For the second round of user tests I created detailed wireframes (Figure 37) of main screens with images of the shop so that users could have a better context for AR part and fixed copywriting and interface issues uncovered earlier.

I gave more concrete tasks to women:

- You walk in the shop and on the first mannequin you see the dress you like. You want to know what reviews women left about it.
- You liked the purple dress with a bird pattern, but it was too small in your chest. You want to leave a review.

Results

- 5 women considered rating to be a mood selection
- 4 women admitted that they wanted to write helpful reviews but had no idea how to do it
- 3 women could not find how to switch from map to shop view

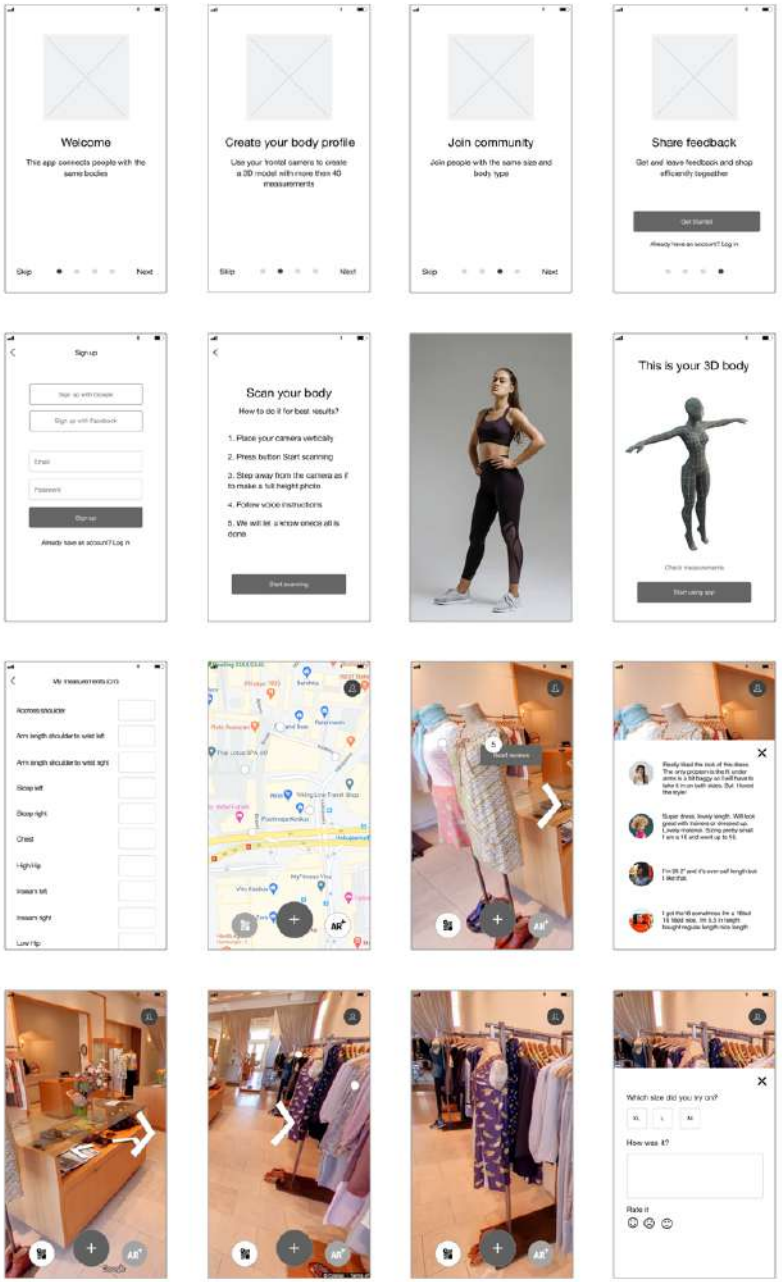


Figure 37: Wireframes of the main screens

- 2 women had questions regarding such details as how to place phone so that it could scan a body, although creating a body scan process was clear to them

Prototyping - round 3

For the third round of user tests I created fully designed screens and created a prototype in Figma. I switched to Figma from Invision because I needed a function of automatic transition between screens, but Invision did not support it.

During previous user testings I tested key functions, but for this round I:

- defined the look of the whole app. I made it neutral and used friendly neutral colours, because potential users of my solution have similar problems, but they have different tastes.
- added additional functions, such as reactions to the comments, viewing reviews on the map, welcome screen, profiles
- brought out women faces more by introducing galleries of looks created by shoppers.

AR prototype

I still had lots of questions about how to see and add comments in augmented reality:

- Should reviews be blended into the digital world or be part of a smartphone interface?
- If UI elements are blended in, how will a user interact with them?

I created an AR prototype in Torch (Figure 39) to find answers to those questions. I used Image tracking function of the app and my clothes stand. I chose patterns (Figure 38) that will be recognized, and once they are in focus, reviews indicators would appear.



Figure 38: Defining patterns tracked in Torch



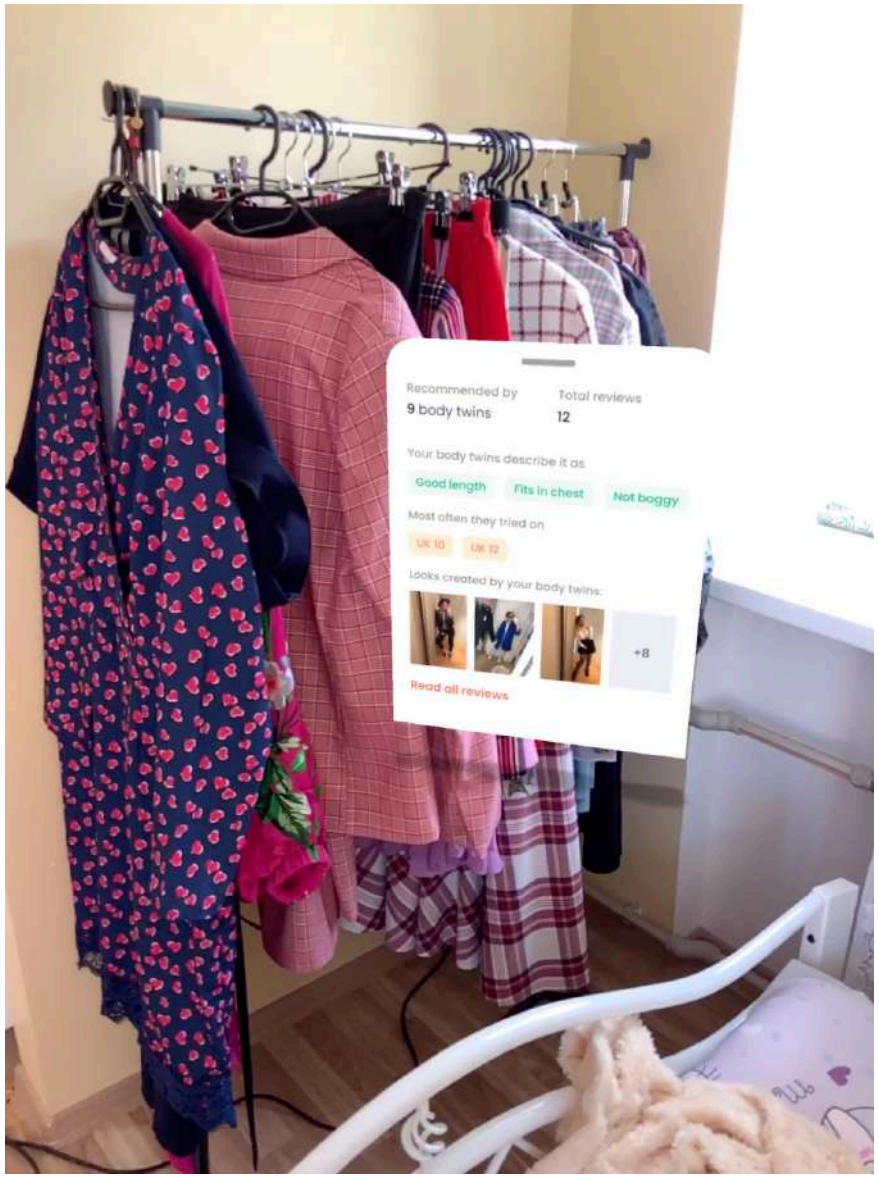


Figure 39: Prototype made in Torch

Key findings

While testing AR prototype, I found out that having all reviews and comments blended into the real world is not very convenient because the phone should always be focused on the same place. Considering that the solution is used in a public place, such interaction with the device is not socially acceptable: shop assistance can be against making a video in the shop or people around can think that a person is playing Pokemon Go or just being weird (Figure 40).

I decided that the best way to use my solution is to be able to see reviews indicators in AR as blended into the surroundings, but open and read them in a way we scroll Instagram feed.

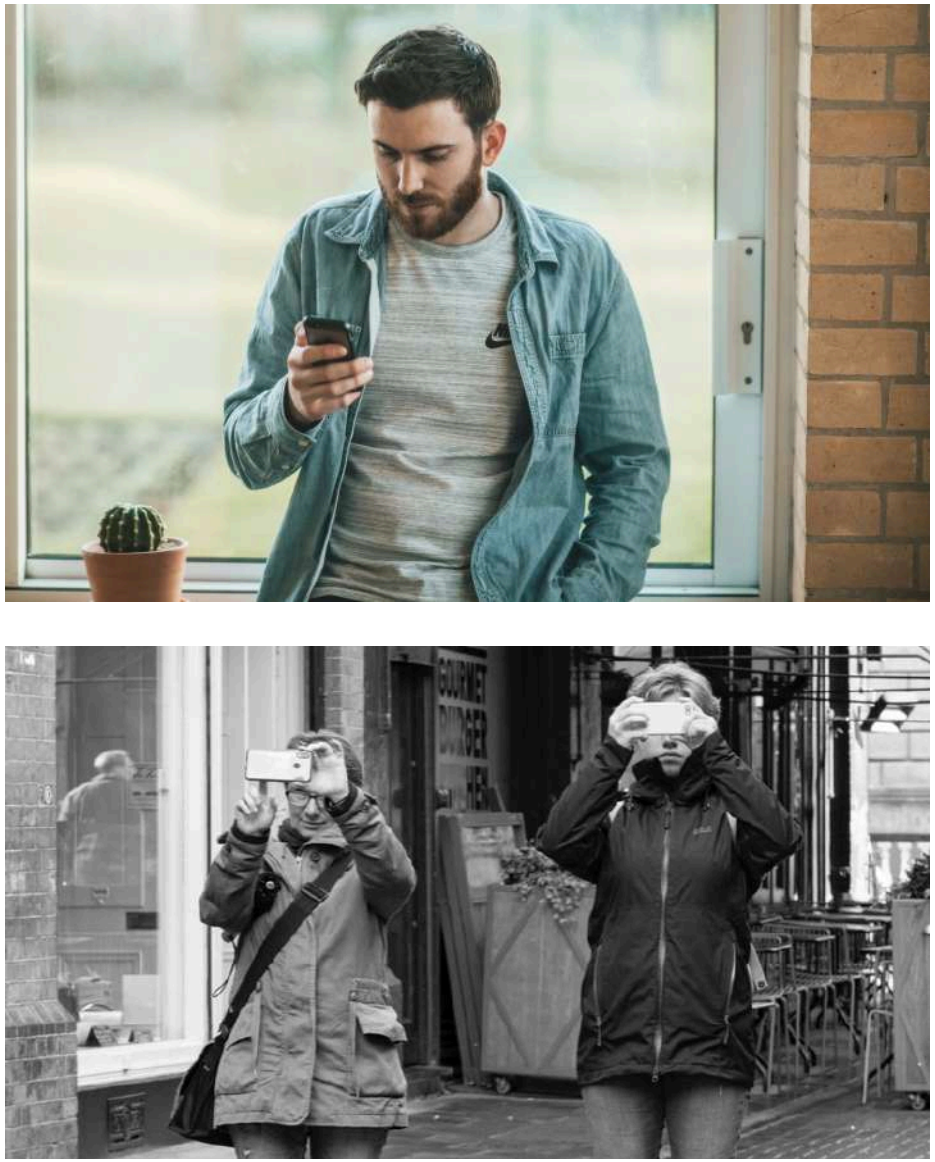


Figure 40: Connotation behind smartphone interaction, unsplash.com

Testing - round 3

I tested my solution using the same scenarios I had in the previous round, but had longer follow up interviews to find out what are the general impressions on my solutions, what women liked or disliked about it, how logical was the experience. I invited people I interviewed for my research and people from the same Telegram group described previously. For those who I interviewed previously I customized the screens so that they could feel the context better and relate to them by using their Facebook avatars.

I mostly got positive comments and also interesting suggestions, that could get further development.

Body scanning

'I mostly don't know my sizes and don't remember the last time I measured something. So I think it is good that you have a model and then can check.'

Map and filtering

'I would put shorts separately from jeans and trousers. Who wears shorts? I think it is a very specific item.'

'I don't have skirts. The ones of my size are mini. Maybe I could finally find something?'

Reviews

'I never read reviews on websites but it turned out to be very helpful. People usually write if size is bigger or smaller than stated. It is good that the app has it as well'

'I like that there is a summary of reviews. I never read reviews but this is a timesaver'

Meeting a body twin

'What if you go shopping with you body twin and you like the same dress? But on the other hand all people have different tastes, and she might show you something you would never try on otherwise.'

Key points I managed to achieve were:

- making reviews useful for both readers of the reviews and those who don't read them,

- finding text suggestions, that were important to women. All women said that knowing if a size is smaller or bigger then stated was important.

Final improvements

I managed to fix most of the issues, uncovered earlier, except the on-boarding. In my opinion it became a recurrent problem because the pattern of map and pin selection and the pattern of using AR are known to users separately, but when they are combined, it creates confusion. To make the process of using the app more understandable in the final version I decided to add instructions to the main elements of the screens.

Adding tooltips (Figure 41) also influenced the design of the tab bar. Tooltips introduce main functions, adding a review button didn't have to be that big and prominent any more. I reduced its size, and also added notifications and other functions to the same place, to have them less scattered around the screen (which was an issue during the previous test rounds) and make them more prominent on the white background. Also, this design of the tab bar would enable user access adding a review function from any screen of the app.

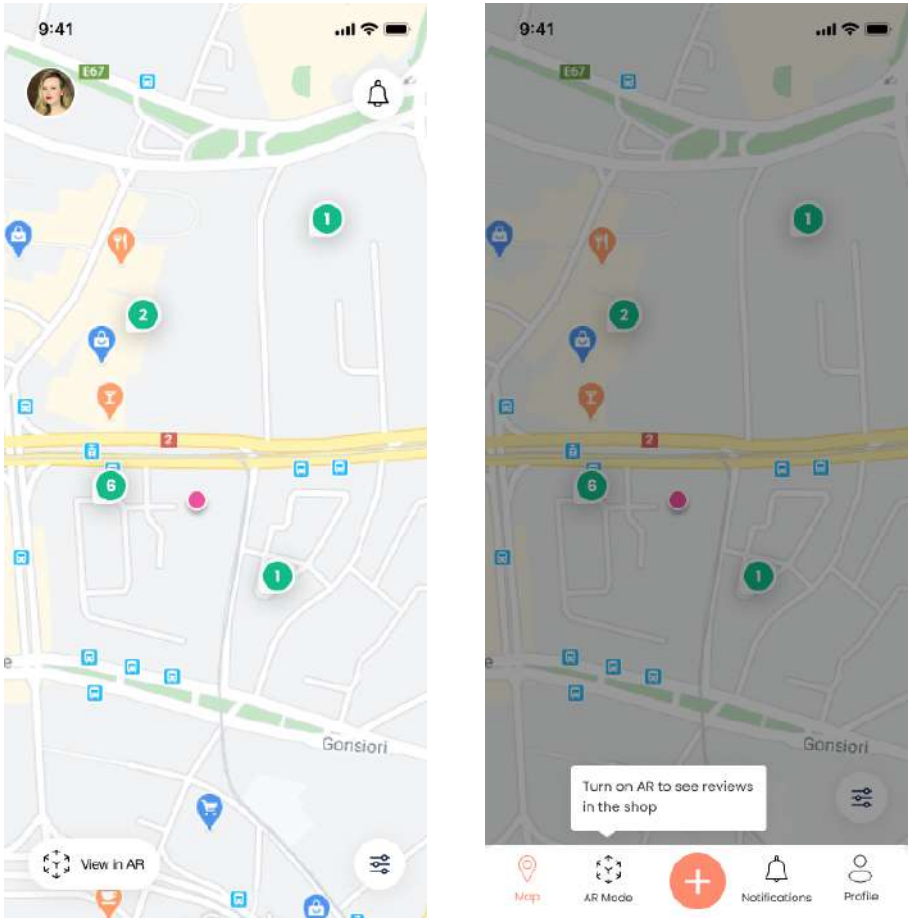


Figure 41: Variant 3 and final version

8. Final solution

8.1 Design component

BodyTwins is an app which offers an alternative way of shopping by showing relevant reviews in real-time context through augmented reality. It connects women having the same body parameters in a network of contributors.

Link to the prototype:

https://bit.ly/3bWTPG4

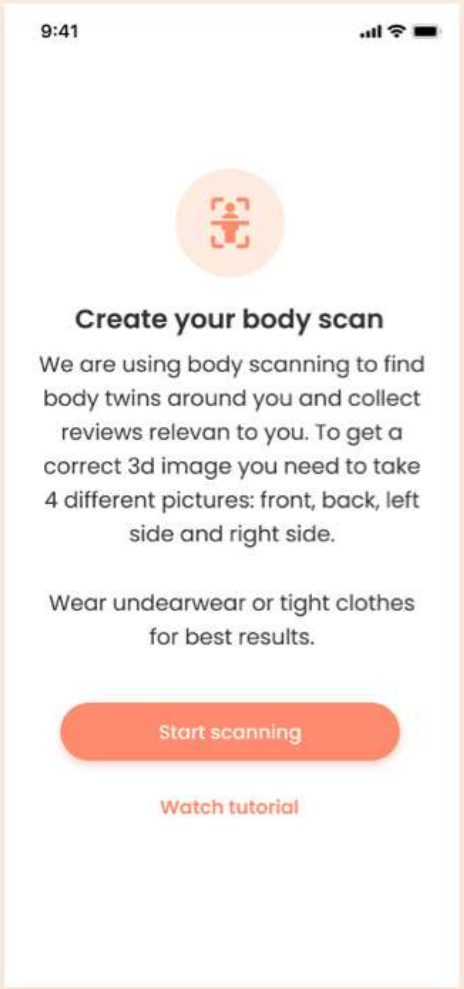
Link to the video:

https://vimeo.com/user66347655

Body scanning

Kate fills in basic registration about herself or logs in using either email or Instagram and gets to the process of body-scanning. Body is scanned using time of flight sensor. To do this, she needs to place her phone vertically and follow voice instructions on whether she is in the right position or she needs to turn. She can also see the tutorial on how to do it correctly before starting the process.

I introduced registration through Instagram to encourage women follow their body twins there and thus influence creation of a more realistic instagram feed.



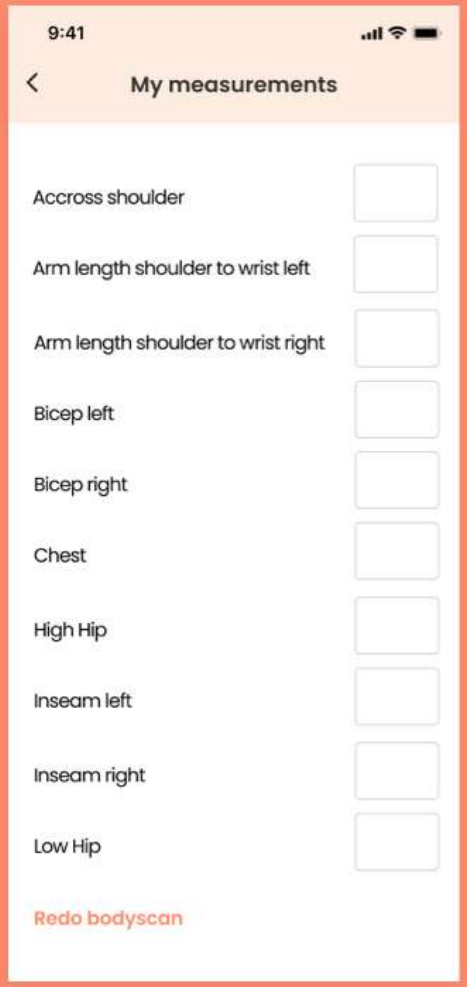


When the scan is done, voice instruction notifies her about it, and she can see her body in 3D. She can rotate and zoom in and out the created model.

User testings showed that some women doubted if the 3D model is theirs, that's why I added an option to see the measurements and adjust them if needed.

A user can re-do her body scan immediately or be reminded to re-do it over time by a push notification. This option is also always available in account settings.





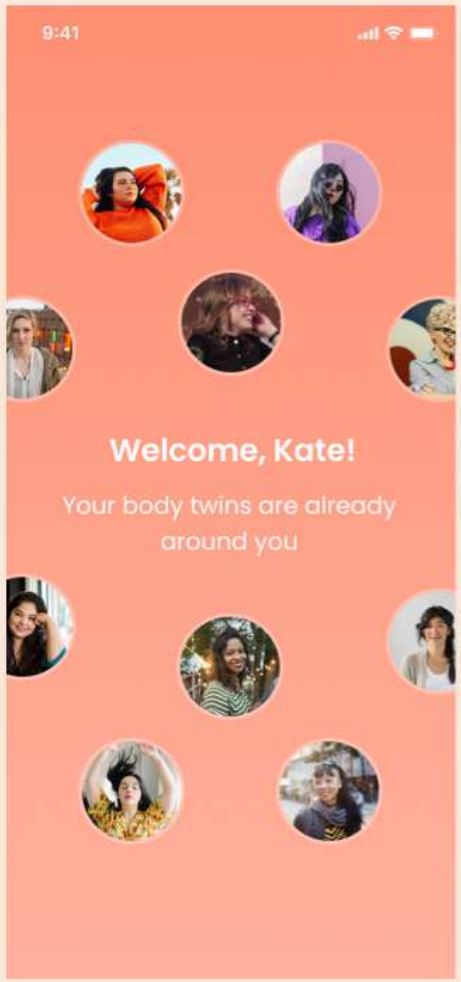
Welcome screens

Kate finished registration and sees a welcome screen, that shows avatars of the people who are her body twins and are using this app. This screen serves as an introduction of what she will see next.

Kate used to say:

'I try to look for people like me. Sometimes I ask my friends: 'Does this girl look like me? I am not that bad.'

Now she sees that there are women like her around.



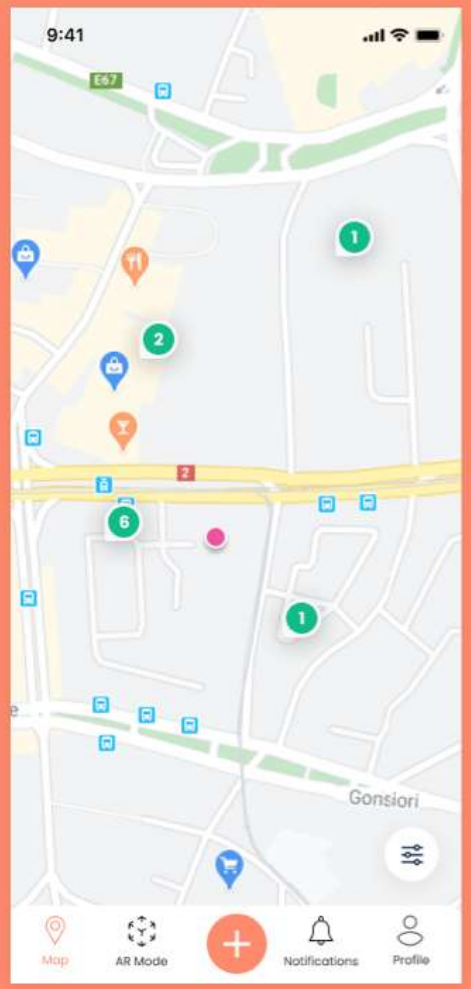


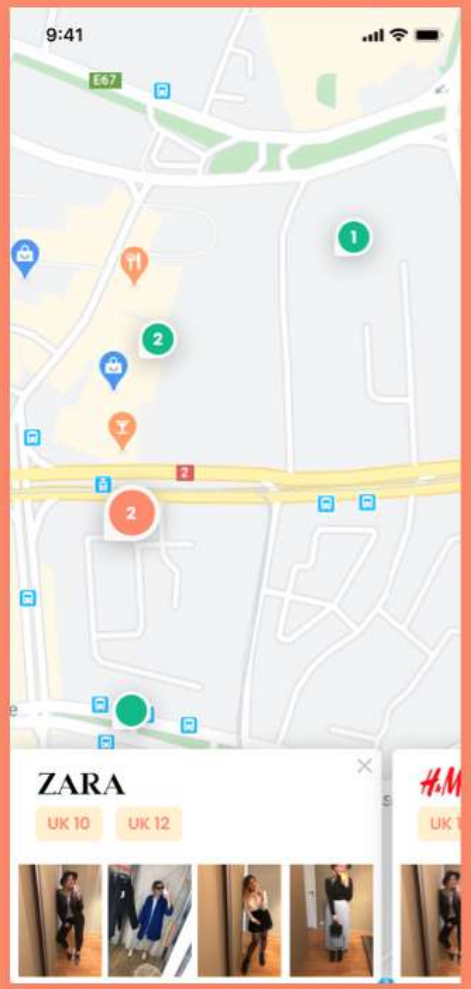
Map view

Kate is at home. She is having her business trip abroad soon and is planning what clothes she can take with her. She can not wear leggings or anything from kid's department because she is going to meet partners and needs to look professional. She realizes that she does not have summer trousers, but she doesn't want to go shopping. Last time she needed a new dress for New Year corporate event, and she ended up crying in a dressing room because everything was too long or too big.

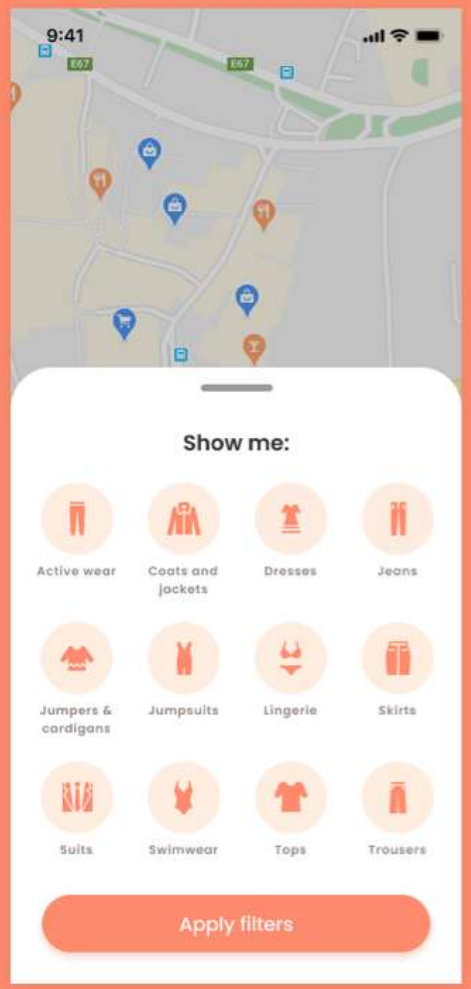
Kate used to think: 'When I go shopping, I expect that most likely I won't find anything.'

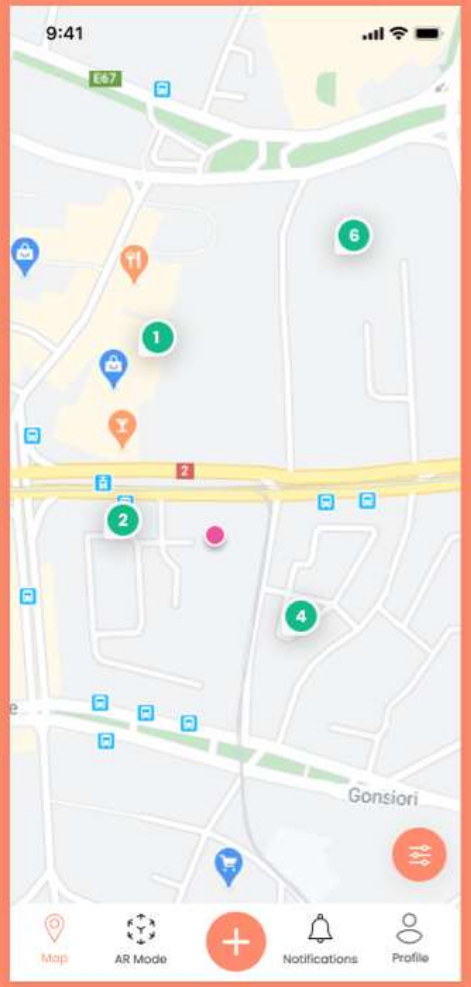
Now she looks at the map and sees that women have already left positive reviews on the clothes. So there is hope for her. She can read a summary about clothes: which sizes women tried on, how they describe them and what looks they made with those clothes. Now she can get an idea of what she can buy and decides to go shopping.





She can also filter reviews by main categories to check if there is something she can wear. For example, she can see if someone found trousers in XXS size.



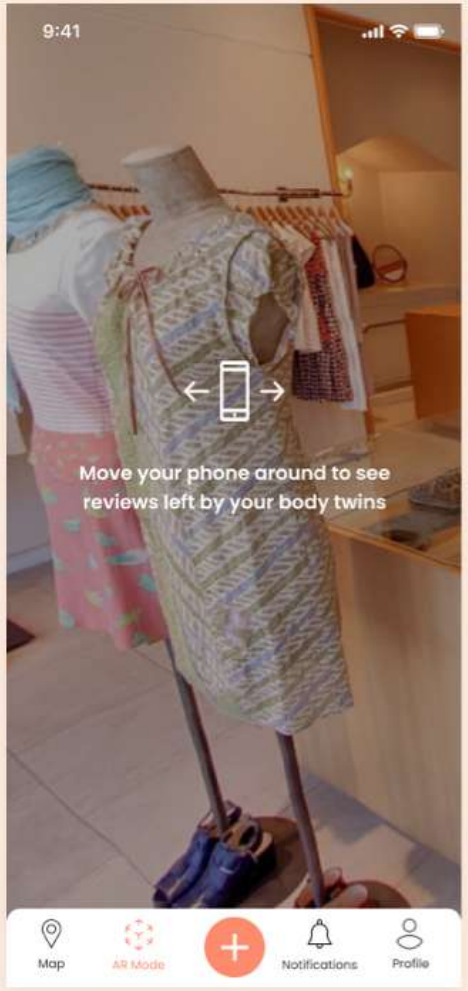


Reviews in augmented reality

Kate goes to the shop and turns on AR mode of the app in one of the shops. It starts with an instruction on how to use her phone in AR mode. She quickly looks around through the screen and sees all the clothes with positive and negative reviews.

Each label is marked with colour, red or green, indicating if the majority of reviews are positive or negative. The closer Kate is to the label, the bigger and prominent it is. The nearest ones are shown as full buttons with labels to serve as an additional hint that they can be clicked. It is also a way to instruct new users on how to use the app.

If the clothes are invisible from her angle or are in a different section, there are arrow indicators with numbers showing her direction. This feature is introduced to point out clothes in unexpected locations: for example, kid's or men's departments.





Kate used to say:

‘In the shop I see that there is a variety of choice for everyone. But this choice ends on me’.

Kate sees that her body twins have the same problems, lots of clothes don’t fit them properly. This way she can relate to them and realize that she is not alone with her problem.





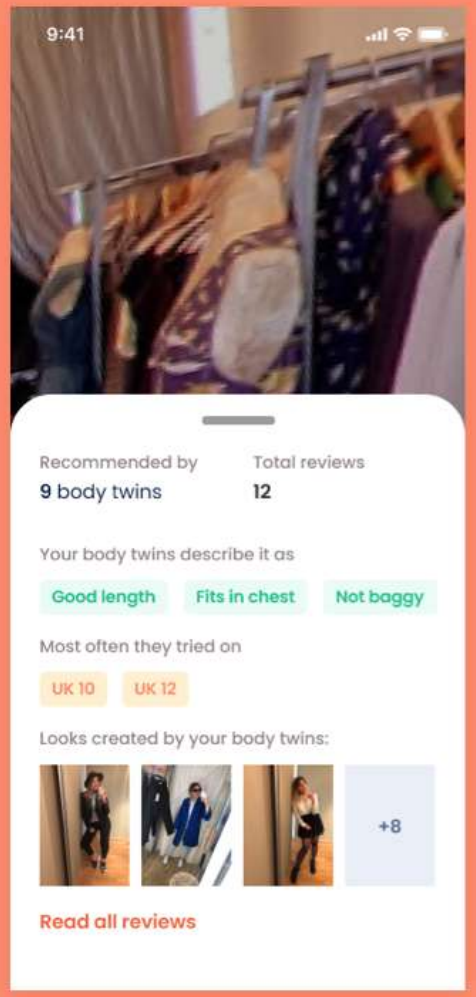
Review details

Kate likes a dress nearby, and she reads reviews added by her body twins. She finds a summary with sizes tried on, main clothes descriptions and looks with those clothes.

Kate used to say:

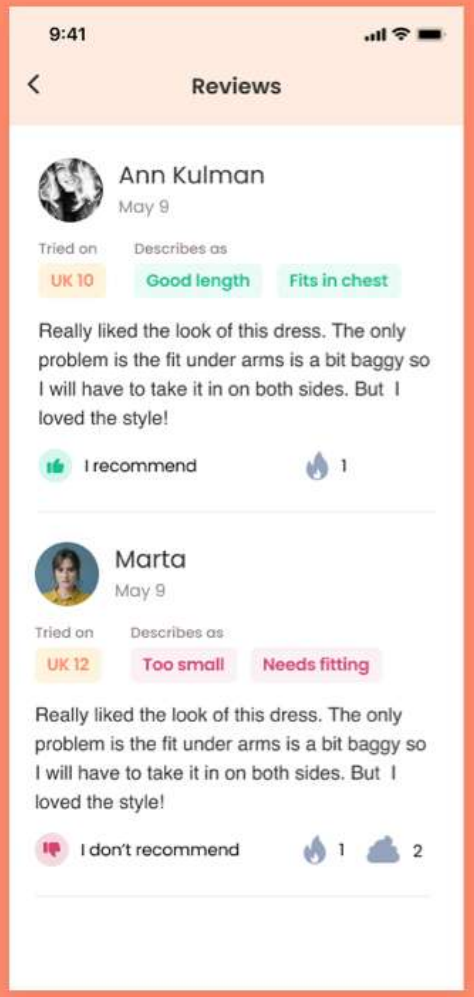
‘Now I can see clothes on models of my size, but not my height. Maybe on the website dwarf.com'.

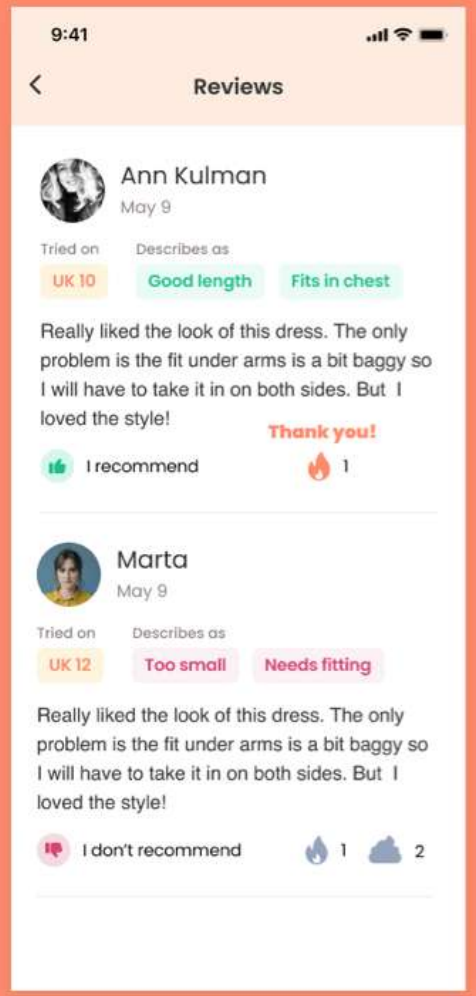
Now she can see clothes on people like her and clearly understand what they would look like on her.





She can also read all reviews left by women. Kate can react to reviews by pressing 'Fire' to show that the review was very helpful to her, and 'Poop' to let a reviewer know that same happened to her.

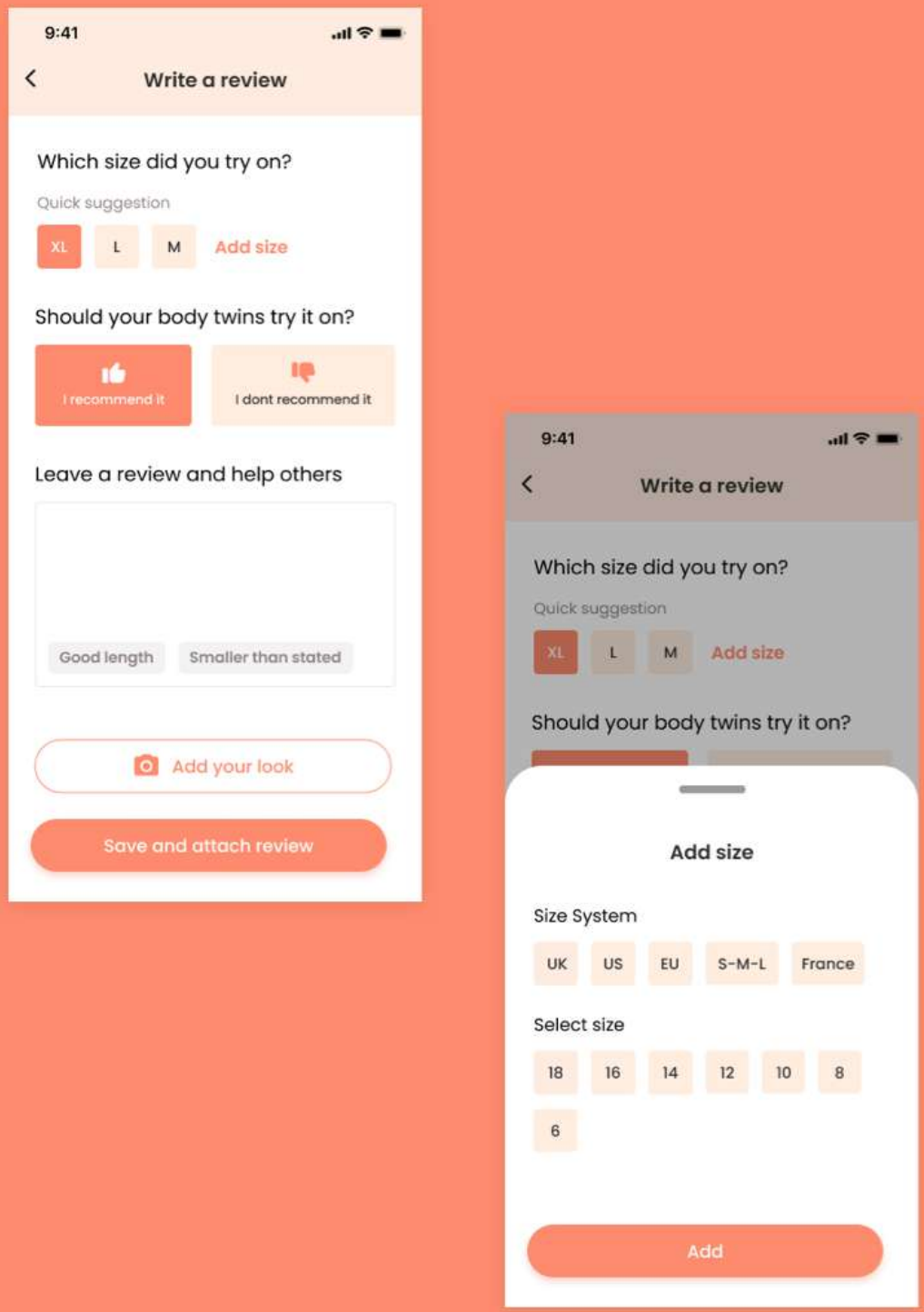




Posting a review

Although Kate went shopping for trousers, she considers other options. Usually, she takes a big pile of clothes and tries everything on, but this time her pile is a bit smaller. This time Kate doesn't take clothes already tried on by other women, but the ones without virtual labels. She goes to a dressing room, tries them on as usual. Some clothes fit fine, and some don't. She believes that sharing is caring and sees that so many women helped her with their reviews. That's why she decides to post reviews on the clothes she tried on as well.

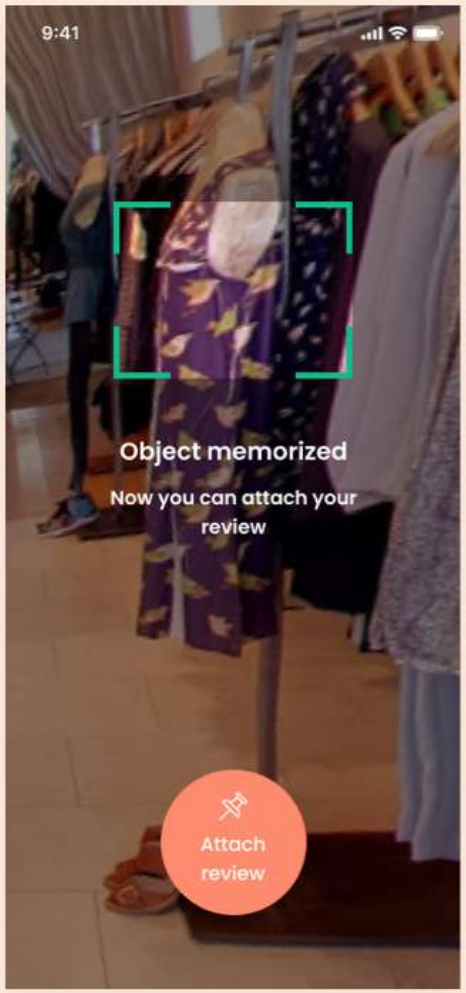
She can select a size that she tried on, add a description and a picture. Because app knows her measurements, it suggest sizes that she would probably choose to speed up the process of leaving a review, but she can also add a size herself. The app offers text suggestions on how to describe the clothes to make reviews more helpful. The suggestions are based on Kate's measurements, research and words from other people's reviews.



When the review is ready, she can attach it to the item she had tried on.

'I know certain brand that will definitely fit me. For example Levi's have size 24 jeans. Even 23.'

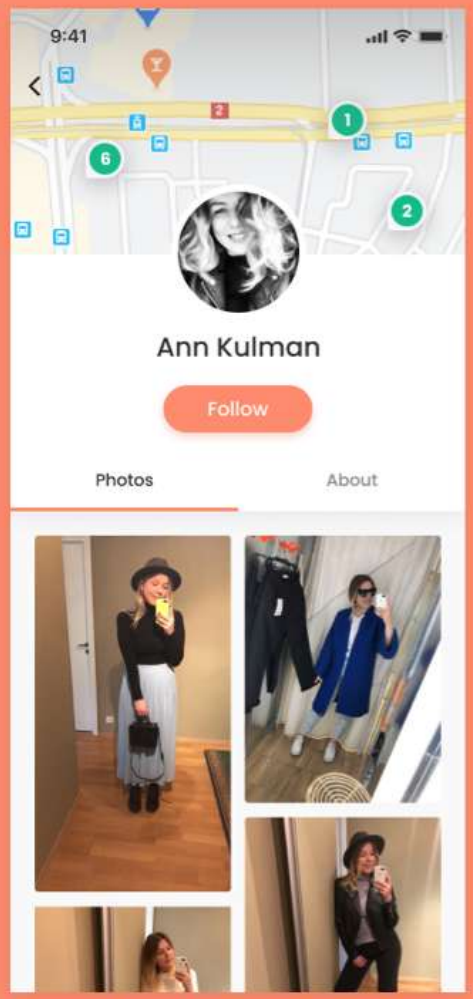
Now she can share this knowledge with her body twins.

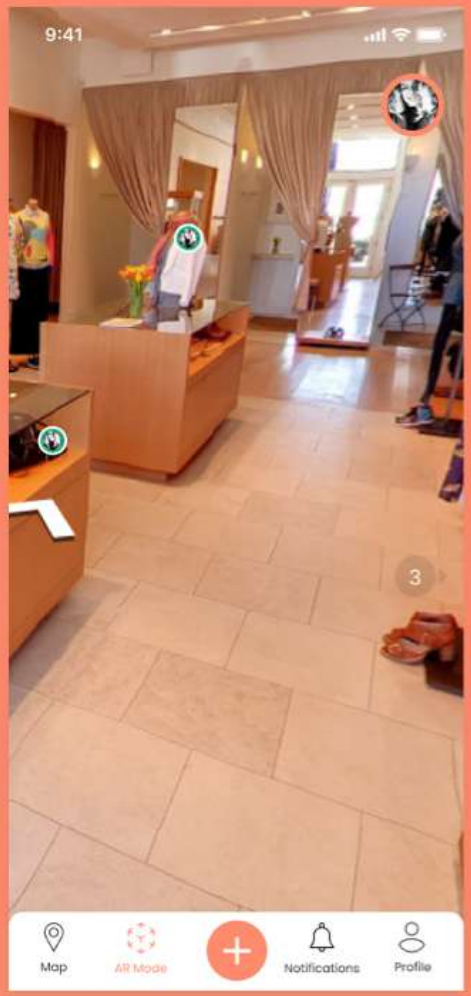




Following other people

Kate uses the app very often and started noticing that she mostly likes looks and reviews added by the same person. So she decides to follow Ann. Kate can also see the link to her Instagram account and start following her there as well. Now when she uses her AR mode, she can click on Ann's avatar and see reviews only by her.





Meeting your body twin

Sometimes Kate can not decide if she likes clothes or not, and she is not sure if that's the way they should look on her. Shop assistants are not very helpful because they want to sell their clothes and don't really know how they are supposed to look on different body types.

There is a special function in the app for this that Kate can activate if she wants to. If her body twin is also in the shopping mall, she will receive a notification about it. And they can meet and go shopping together.



8.2 Psychology behind the solution

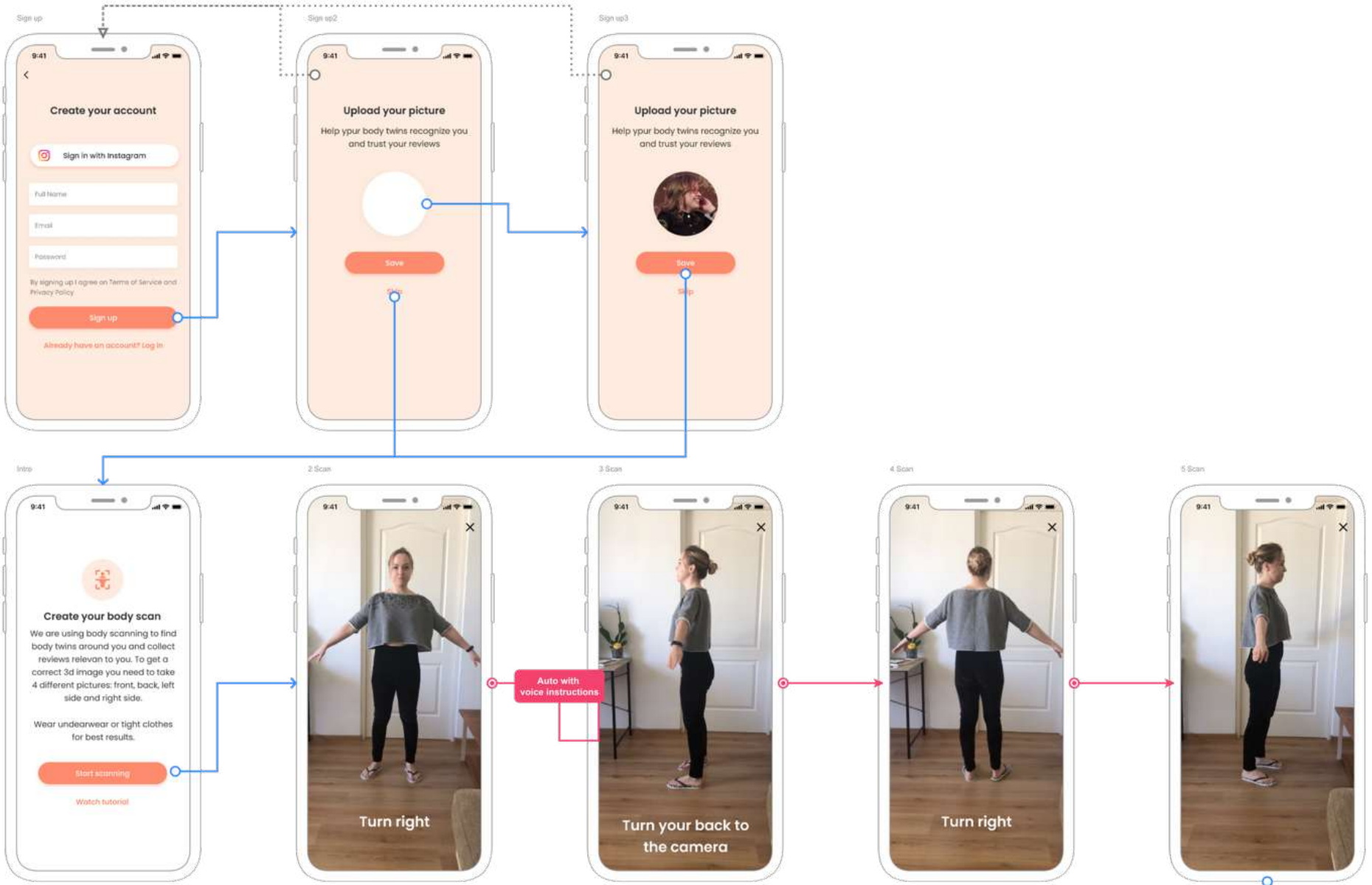
If we have a look at the app from **Shame Resilience Theory**, we can see that my solution provides several ways to build shame resilience.

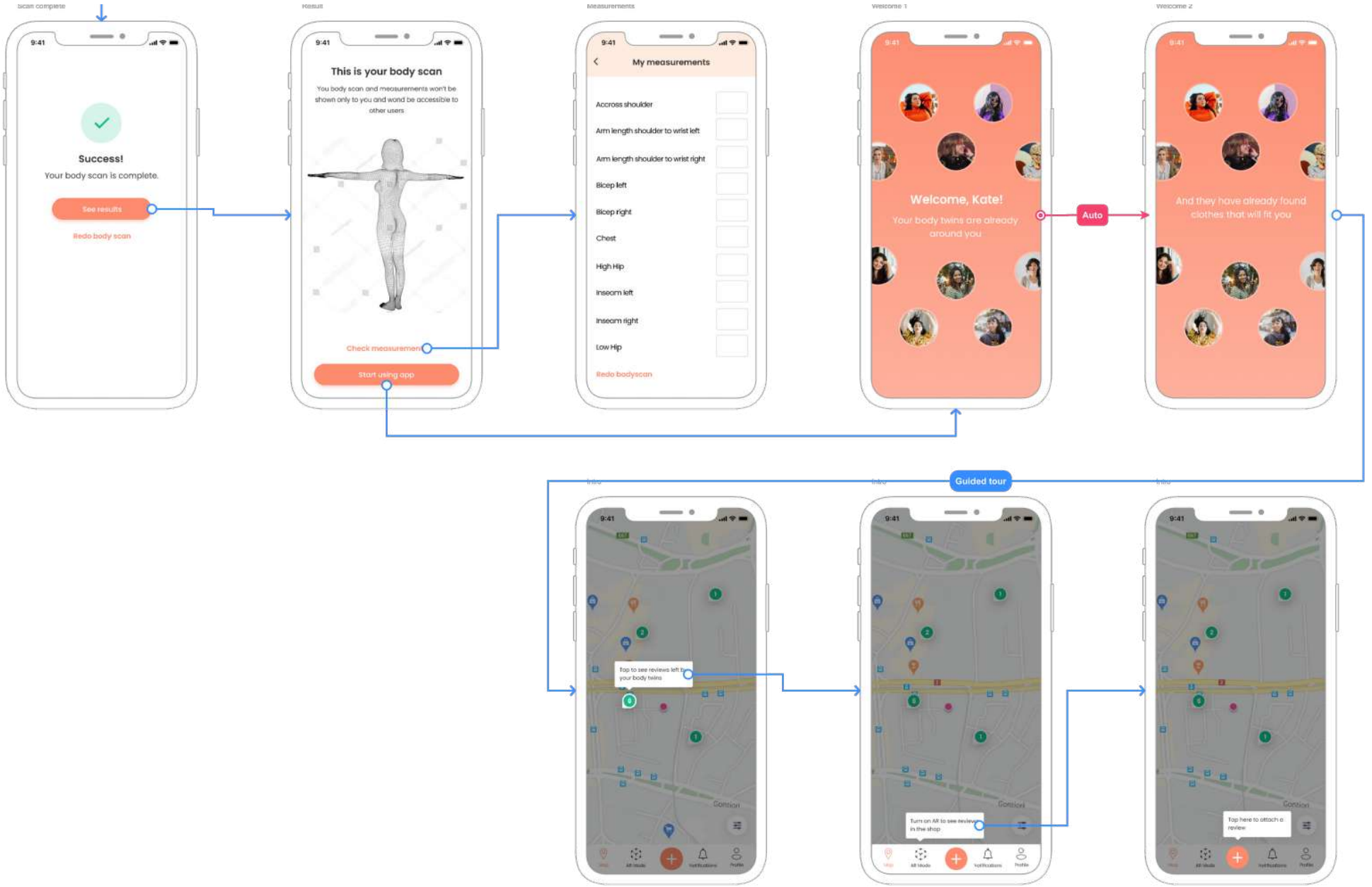
My whole solution is aimed at shopping as a body dissatisfaction trigger, and because of its purpose it already serves as a way of **acknowledging personal vulnerability**. In other words, if a woman downloads an app, it is already an unconscious first step to addressing her trigger.

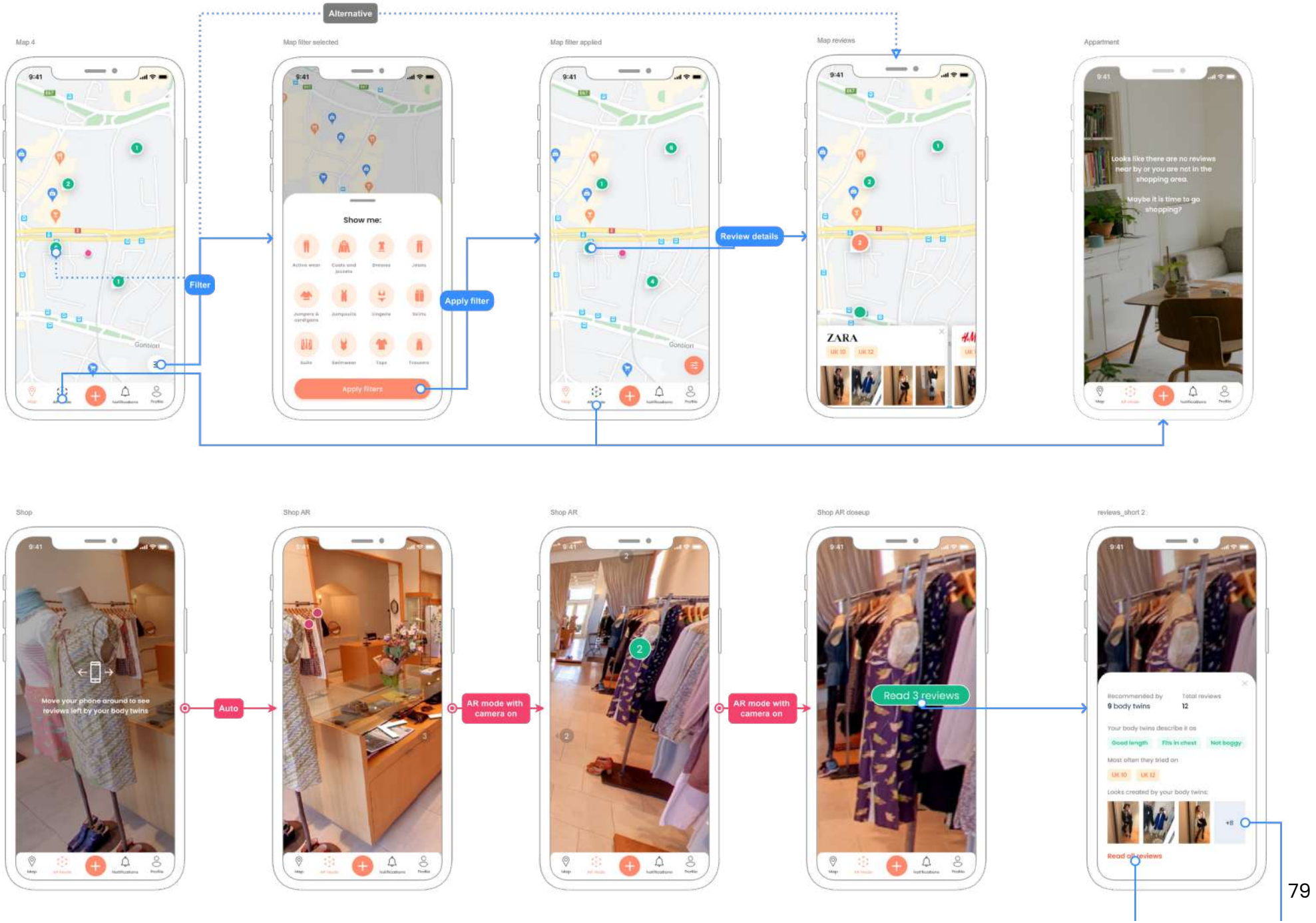
On the level of **practicing critical awareness**, the solution does not impose an ideal of how a woman should look, but brings out real people, usually invisible in media. By showing real women it creates a way to normalize real bodies in real situations. It provides a bigger context for feeling shame while shopping for clothes. As Brene Brown says, 'shame works only if we are alone in it. If we think there is someone else, a group of women, a world full of women, struggling with the same issue, the concept of shame becomes bankrupt.'

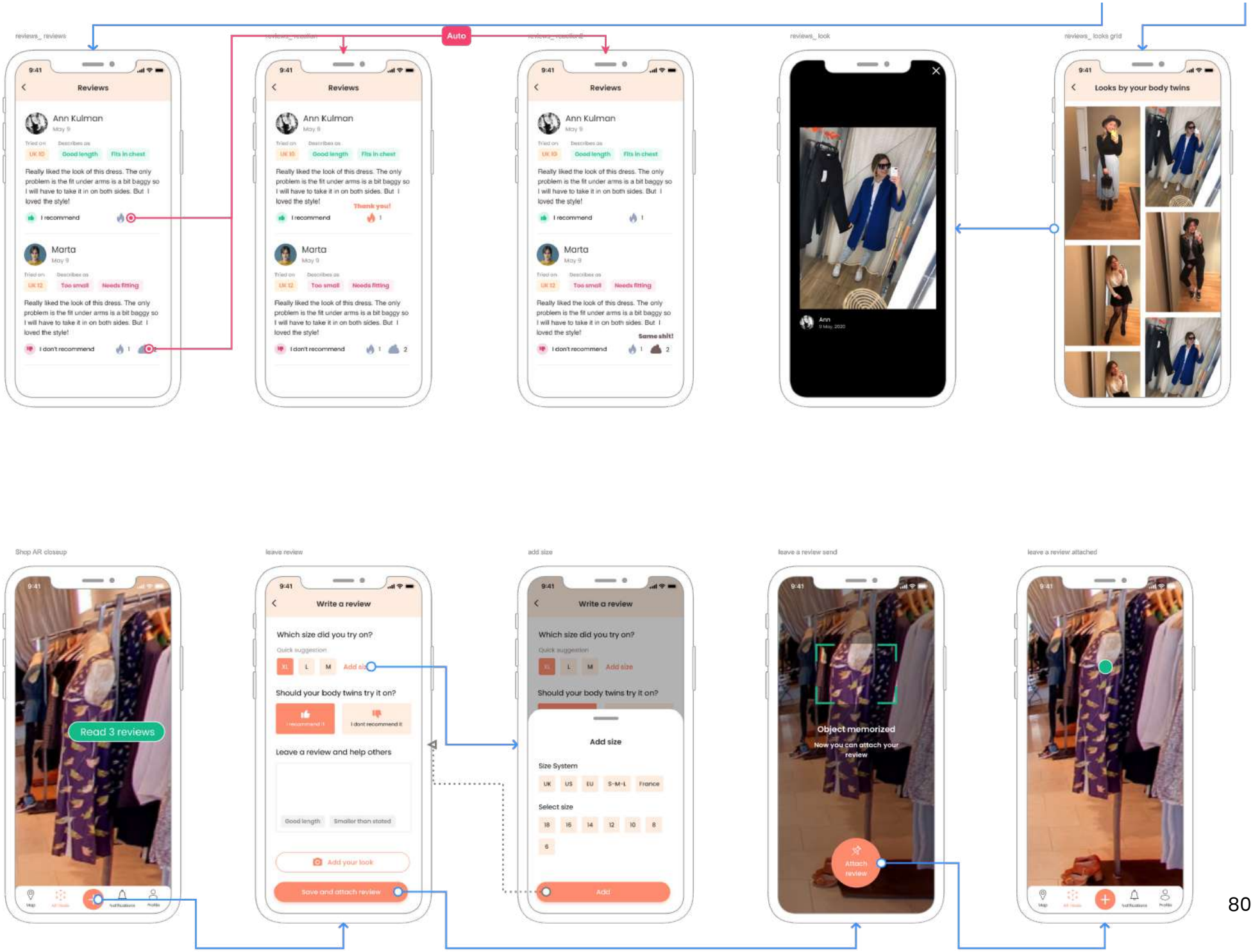
Reaching out manifests itself in the solution in the form of reviews and reactions, which are a way to share a personal story within a context of shopping.

8.3 Main flows




Success!
Your body scan is complete.
See results
Redo body scan
This is your body scan
You body scan and measurements won't be shown only to you and wond be accessible to other users
Check measurements
Start using app
My measurements
Accross shoulder
Arm length shoulder to wrist left
Arm length shoulder to wrist right
Bicep left
Bicep right
Chest
High Hip
Inseam left
Inseam right
Low Hip
Redo bodyscan
Welcome, Kate!
Your body twins are already around you
Auto
And they have already found clothes that will fit you
Guided tour


Alternative
Map 4
Map filter selected
Map filter applied
Map reviews
Appartment
Filter
Show me:
Apply filters
Apply filter
Review details
ZARA
Looks like there are no reviews near by or you are not in the shopping area.
Maybe it is time to go shopping?
Shop
Shop AR
Shop AR
Shop AR closeup
reviews_short 2
Move your phone around to see reviews left by your body twins
Auto
AR mode with camera on
AR mode with camera on
Read 3 reviews
Recommended by
9 body twins
Total reviews
12
Your body twins describe it as
Good length
Fits in chest
Not boggy
Most often they tried on
UK 10
UK 12
Looks created by your body twins:
+8
Read all reviews


reviews_ reviews
Auto
reviews_ look
reviews_ looks grid
Reviews
Ann Kulman
Marta
Looks by your body twins
Shop AR closeup
leave review
add size
leave a review send
leave a review attached
Read 3 reviews
Write a review
Which size did you try on?
Should your body twins try it on?
Leave a review and help others
Add your look
Save and attach review
Add size
Size System
Select size
Add
Object memorized
Now you can attach your review
Attach review

9. Reflections

9.1 Process reflection

In my opinion, carrying out this thesis project has given a boost to me as a designer and added valuable skills and attitudes to my future work.

Firstly, I managed to find efficient strategies of narrowing down and finding the right focus through choosing the right methods at the right time. It was especially important at the beginning, because the topic of body confidence is complex and consists of multiple levels.

Secondly, I fully achieved my personal goals, stated in the beginning of the project: improved my skills of interviewing people and analyzing interviews and got new skills in user testing with professional tools.

Thirdly, I overcame my personal biases and underestimating the power of research. Before the project it felt to me that it means researching all the creativity out of design, but now I can see the value of it and, what is more important, approach more critically my work and works of others, and see and evaluate the amount of research applied in other people's solutions.

Lastly, I mastered my time planning skills and learned a valuable lesson of meeting deadlines.

9.2 Future development

Despite reaching my goals for this project, I would like to refine my solution further. I plan to test my solution in a set up similar to the shop environment, which was impossible because of coronavirus. Although I tested functions of my solution and got feedback on it, I feel that testing in real environment can give a better understanding of psychology behind the solution. How women would feel if they saw their body twins? Would it be uncomfortable? Would their attitude to their body change over time?

In addition to that, I would like to scale my solution to AR glasses and see how change of the device will influence interaction. Probably lot's of interactions will happen just by

having an object in focus and won't require any gestures. How could information be shown while a user is walking among the displays? What can they see when they decide to have a closer look at the clothes?
This set of questions opens new opportunities for further development of the solution.

9.3 Solution within a context of bigger trends

Shopping after corona

Lifestyle and shopping will change after coronavirus, especially in-store shopping, which my solution is aimed at. On the one hand, Report by the Business of Fashion (BoF) and consultants McKinsey products states that global fashion sales are predicted to fall by up to 30% in 2020, and the luxury end of the market will be even harder hit, with sales slumping by up to 40%[28]. It also predicts that social distancing has already made digital an urgent priority. On the other hand, people do not always act logically, and we can not fully predict their behaviour. For example, Hermes's flagship store in Guangzhou recorded US$2.7 million in sales (S$3.8 million) just a day after reopening [29], due to "revenge spending" (a buying binge by shoppers emerging from lockdown). My solution can still be viable within both scenarios: either if women are happy and enjoy coming back to normal life more than ever, or if they need a way to minimize time spent in crowded places and a tool for faster decisions making when shopping.

Search for meaningful connection

GlobalWebIndex's Consumer Trends for 2019 report suggests that stepping away or downsizing on social media has become an increasing trend throughout the past year. According to, Reddit Co-founder Alexis Ohanian, people want to belong to a smaller community, where they can be heard and where they can meet like-minded people. "What people seem to be clamoring for more and more now is community, and so whether that goes to private group chats — like your WhatsApp group, your Signal group — or new platforms that have emerged," he said [30]. My solution can provide an active

experience around the subject of interest for people interested in fashion and those who need help and support.

Overcoming insecurities

Size and body shape is a topic widely discussed in media and among people, but rarely spoken out loud in a context of everyday activities. As one of the women I interviewed said: 'I like sauna, but I don't go there. Everyone around me will be slim and sporty, and I have a fat belly.' People think about it and avoid many pleasurable activities, where their body will be exposed. I believe my solution can be implemented in different ways: help people overcome their insecurities by organizing interest-based groups: go together to the sauna, start working out with people like you or organize an efficient clothes swap party and be sure that everyone will find something they like.

Feminism

I started my thesis with a quote by Naomi Wolf, and would like to finish it with another Naomi Wolf's quote: 'Ironically, the myth that drives women apart also binds them together. Commiserating about the myth is as good as a baby to bring strange women into pleasant contact, and break down the line of Other Woman wariness. A wry smile about calories, a complaint about one's hair, can evaporate the sullen examination of a rival in the fluorescent light of a ladies' room. On one hand, women are trained to be competitors against all others for "beauty"; on the other, when one woman—a bride, a shopper in a boutique—needs to be adorned for a big occasion, other women swoop and bustle around her in generous concentration in a team formation as effortlessly choreographed as a football play. These sweet and satisfying rituals of being all on the same side, these all-too-infrequent celebrations of shared femaleness, are some of the few shared female rituals left; hence their loveliness and power. But, sadly, these delightful bonds too often dissolve when the women reenter public space and resume their isolated, unequal, mutually threatening, jealously guarded "beauty" status.' I believe that from feminist perspective my solution can be a digital way of practicing female culture and create invisible bonds and solidarity between women.

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